

| LOC | OBJECT CODE | ADDR1 | ADDR2 | STMT                                                                     |
|-----|-------------|-------|-------|--------------------------------------------------------------------------|
|     |             |       |       | 2 *****                                                                  |
|     |             |       |       | 3 *                                                                      |
|     |             |       |       | 4 * Concurrent-functions facility tests                                  |
|     |             |       |       | 5 * for SSF encoded instructions:                                        |
|     |             |       |       | 6 *                                                                      |
|     |             |       |       | 7 * C8x6 CAL - Compare And Load                                          |
|     |             |       |       | 8 * C8x7 CALG - Compare And Load Long                                    |
|     |             |       |       | 9 * C8xF CALGF - Compare And Load Long Fullword                          |
|     |             |       |       | 10 *                                                                     |
|     |             |       |       | 11 * James Wekel March 2026                                              |
|     |             |       |       | 12 *****                                                                 |
|     |             |       |       | 13                                                                       |
|     |             |       |       | 14 *****                                                                 |
|     |             |       |       | 15 *                                                                     |
|     |             |       |       | 16 * basic instruction tests                                             |
|     |             |       |       | 17 *                                                                     |
|     |             |       |       | 18 *****                                                                 |
|     |             |       |       | 19 * This program tests proper functioning of the z/arch                 |
|     |             |       |       | 20 * Concurrent-functions facility compare and load instructions         |
|     |             |       |       | 21 * (CAL, CALG, CALGF). Exceptions are not tested.                      |
|     |             |       |       | 22 *                                                                     |
|     |             |       |       | 23 * PLEASE NOTE that the tests are very SIMPLE TESTS designed to catch  |
|     |             |       |       | 24 * obvious coding errors. None of the tests are thorough. They are     |
|     |             |       |       | 25 * NOT designed to test all aspects of any of the instructions.        |
|     |             |       |       | 26 *                                                                     |
|     |             |       |       | 27 *****                                                                 |
|     |             |       |       | 28 *                                                                     |
|     |             |       |       | 29 * *Testcase cff-01-CompareAndLoad                                     |
|     |             |       |       | 30 * *                                                                   |
|     |             |       |       | 31 * * Concurrent-functions facility tests for SSF encoded instructions: |
|     |             |       |       | 32 * *                                                                   |
|     |             |       |       | 33 * * C8x6 CAL - Compare And Load                                       |
|     |             |       |       | 34 * * C8x7 CALG - Compare And Load Long                                 |
|     |             |       |       | 35 * * C8xF CALGF - Compare And Load Long Fullword                       |
|     |             |       |       | 36 * *                                                                   |
|     |             |       |       | 37 * * # -----                                                           |
|     |             |       |       | 38 * * # This tests only the basic function of the instruction.          |
|     |             |       |       | 39 * * # Exceptions are NOT tested.                                      |
|     |             |       |       | 40 * * # -----                                                           |
|     |             |       |       | 41 * *                                                                   |
|     |             |       |       | 42 * mainsize 2                                                          |
|     |             |       |       | 43 * numcpu 1                                                            |
|     |             |       |       | 44 * sysclear                                                            |
|     |             |       |       | 45 * archlvl z/Arch                                                      |
|     |             |       |       | 46 *                                                                     |
|     |             |       |       | 47 * loadcore "\$(testpath)/cff-01-CompareAndLoad.core" 0x0              |
|     |             |       |       | 48 *                                                                     |
|     |             |       |       | 49 * diag8cmd enable # (needed for messages to Hercules console)         |
|     |             |       |       | 50 * runtest 2                                                           |
|     |             |       |       | 51 * diag8cmd disable # (reset back to default)                          |
|     |             |       |       | 52 *                                                                     |
|     |             |       |       | 53 * *Done                                                               |
|     |             |       |       | 54                                                                       |
|     |             |       |       | 55 *****                                                                 |

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|-----|-------------|-------|-------|---------------------------------------------------------------------|
|     |             |       |       | 57 *****                                                            |
|     |             |       |       | 58 * FCHECK Macro - Is a Facility Bit set?                          |
|     |             |       |       | 59 *                                                                |
|     |             |       |       | 60 * If the facility bit is NOT set, an message is issued and       |
|     |             |       |       | 61 * the test is skipped.                                           |
|     |             |       |       | 62 *                                                                |
|     |             |       |       | 63 * Fcheck uses R0, R1 and R2                                      |
|     |             |       |       | 64 *                                                                |
|     |             |       |       | 65 * eg. FCHECK 134, 'vector-packed-decimal'                        |
|     |             |       |       | 66 *****                                                            |
|     |             |       |       | 67 MACRO                                                            |
|     |             |       |       | 68 FCHECK &BITNO, &NOTSETMSG                                        |
|     |             |       |       | 69 .* &BITNO : facility bit number to check                         |
|     |             |       |       | 70 .* &NOTSETMSG : 'facility name'                                  |
|     |             |       |       | 71 LCLA &FBBYTE Facility bit in Byte                                |
|     |             |       |       | 72 LCLA &FBBIT Facility bit within Byte                             |
|     |             |       |       | 73                                                                  |
|     |             |       |       | 74 LCLA &L(8)                                                       |
|     |             |       |       | 75 &L(1) SetA 128, 64, 32, 16, 8, 4, 2, 1 bit positions within byte |
|     |             |       |       | 76                                                                  |
|     |             |       |       | 77 &FBBYTE SETA &BITNO/8                                            |
|     |             |       |       | 78 &FBBIT SETA &L((&BITNO-(&FBBYTE*8))+1)                           |
|     |             |       |       | 79 .* MNOTE 0, 'checking Bit=&BITNO: FBBYTE=&FBBYTE, FBBIT=&FBBIT'  |
|     |             |       |       | 80                                                                  |
|     |             |       |       | 81 B X&SYSNDX                                                       |
|     |             |       |       | 82 * Fcheck data area                                               |
|     |             |       |       | 83 * skip messgae                                                   |
|     |             |       |       | 84 SKT&SYSNDX DC C' Skipping tests: '                               |
|     |             |       |       | 85 DC C&NOTSETMSG                                                   |
|     |             |       |       | 86 DC C' facility (bit &BITNO) is not installed.'                   |
|     |             |       |       | 87 SKL&SYSNDX EQU *-SKT&SYSNDX                                      |
|     |             |       |       | 88 * facility bits                                                  |
|     |             |       |       | 89 DS FD gap                                                        |
|     |             |       |       | 90 FB&SYSNDX DS 4FD                                                 |
|     |             |       |       | 91 DS FD gap                                                        |
|     |             |       |       | 92 *                                                                |
|     |             |       |       | 93 X&SYSNDX EQU *                                                   |
|     |             |       |       | 94 LA R0, ((X&SYSNDX-FB&SYSNDX)/8)-1                                |
|     |             |       |       | 95 STFLE FB&SYSNDX get facility bits                                |
|     |             |       |       | 96                                                                  |
|     |             |       |       | 97 XGR R0, R0                                                       |
|     |             |       |       | 98 IC R0, FB&SYSNDX+&FBBYTE get fbit byte                           |
|     |             |       |       | 99 N R0, =F'&FBBIT' is bit set?                                     |
|     |             |       |       | 100 BNZ XC&SYSNDX                                                   |
|     |             |       |       | 101 *                                                               |
|     |             |       |       | 102 * facility bit not set, issue message and exit                  |
|     |             |       |       | 103 *                                                               |
|     |             |       |       | 104 LA R0, SKL&SYSNDX message length                                |
|     |             |       |       | 105 LA R1, SKT&SYSNDX message address                               |
|     |             |       |       | 106 BAL R2, MSG                                                     |
|     |             |       |       | 107                                                                 |
|     |             |       |       | 108 B EOJ                                                           |
|     |             |       |       | 109 XC&SYSNDX EQU *                                                 |
|     |             |       |       | 110 MEND                                                            |









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|----------|--------|-----------|----------|----------|---------------------------------|----------------------------------|
|          |        |           |          |          | 235 *****                       |                                  |
|          |        |           |          |          | 236 * cc was not as expecte     |                                  |
|          |        |           |          |          | 237 *****                       |                                  |
|          |        |           | 0000031E | 00000001 | 238 CCMMSG EQU *                |                                  |
|          |        |           |          |          | 239                             |                                  |
|          |        |           |          |          | 240 * extract CC from saved PSW |                                  |
| 0000031E | 5810   | 5058      |          | 00000058 | 241 L R1,CCPSW                  |                                  |
| 00000322 | 8810   | 000C      |          | 0000000C | 242 SRL R1,12                   |                                  |
| 00000326 | 5410   | 831C      |          | 0000051C | 243 N R1,=XL4'3'                |                                  |
| 0000032A | 4210   | 5060      |          | 00000060 | 244 STC R1,CCFOUND              | save actual cc                   |
|          |        |           |          |          | 245                             |                                  |
|          |        |           |          |          | 246 * fill in message           |                                  |
| 0000032E | 4820   | 5004      |          | 00000004 | 247 LH R2,TNUM                  | get test number and convert      |
| 00000332 | 4E20   | 8EBC      |          | 000010BC | 248 CVD R2,DECNUM               |                                  |
| 00000336 | D211   | 8EA6 8E90 | 000010A6 | 00001090 | 249 MVC PRT3,EDIT               |                                  |
| 0000033C | DE11   | 8EA6 8EBC | 000010A6 | 000010BC | 250 ED PRT3,DECNUM              |                                  |
| 00000342 | D202   | 8E4B 8EB3 | 0000104B | 000010B3 | 251 MVC CCPRTNUM(3),PRT3+13     | fill in message with test #      |
| 00000348 | D207   | 8E68 500A | 00001068 | 0000000A | 252 MVC CCPRTNAME,OPNAME        | fill in message with instruction |
|          |        |           |          |          | 253                             |                                  |
|          |        |           |          |          | 254 * fill in expected CC       |                                  |
| 0000034E | B982   | 0022      |          |          | 255 XGR R2,R2                   | get CC as U8                     |
| 00000352 | 4320   | 5007      |          | 00000007 | 256 IC R2,CC                    |                                  |
| 00000356 | 4E20   | 8EBC      |          | 000010BC | 257 CVD R2,DECNUM               | and convert                      |
| 0000035A | D211   | 8EA6 8E90 | 000010A6 | 00001090 | 258 MVC PRT3,EDIT               |                                  |
| 00000360 | DE11   | 8EA6 8EBC | 000010A6 | 000010BC | 259 ED PRT3,DECNUM              |                                  |
| 00000366 | D200   | 8E7E 8EB5 | 0000107E | 000010B5 | 260 MVC CCPRTEXP(1),PRT3+15     | fill in message with CC field    |
|          |        |           |          |          | 261                             |                                  |
|          |        |           |          |          | 262 * fill in found CC          |                                  |
| 0000036C | B982   | 0022      |          |          | 263 XGR R2,R2                   | get CCFFOUND as U8               |
| 00000370 | 4320   | 5060      |          | 00000060 | 264 IC R2,CCFOUND               |                                  |
| 00000374 | 4E20   | 8EBC      |          | 000010BC | 265 CVD R2,DECNUM               | and convert                      |
| 00000378 | D211   | 8EA6 8E90 | 000010A6 | 00001090 | 266 MVC PRT3,EDIT               |                                  |
| 0000037E | DE11   | 8EA6 8EBC | 000010A6 | 000010BC | 267 ED PRT3,DECNUM              |                                  |
| 00000384 | D200   | 8E8E 8EB5 | 0000108E | 000010B5 | 268 MVC CCPRTGOT(1),PRT3+15     | fill in message with CCFFOUND    |
|          |        |           |          |          | 269                             |                                  |
| 0000038A | 4100   | 0055      |          | 00000055 | 270 LA R0,CCPRTLNG              | message length                   |
| 0000038E | 4110   | 8E3B      |          | 0000103B | 271 LA R1,CCPRTLNE              | messagfe address                 |
| 00000392 | 45F0   | 81EC      |          | 000003EC | 272 BAL R15,RPTERROR            |                                  |
|          |        |           |          |          | 273                             |                                  |
| 00000396 | 5800   | 8320      |          | 00000520 | 274 L R0,=F'1'                  | set GLOBAL failed test indicator |
| 0000039A | 5000   | 8E00      |          | 00001000 | 275 ST R0,FAILED                |                                  |
|          |        |           |          |          | 276                             |                                  |
| 0000039E | 47F0   | 8102      |          | 00000302 | 277 B TESTREST                  | continue with test result check  |







[illegible]





| LOC      | OBJECT   | CODE     | ADDR1    | ADDR2    | STMT                                                         |
|----------|----------|----------|----------|----------|--------------------------------------------------------------|
|          |          |          |          |          | 418 *=====                                                   |
|          |          |          |          |          | 419 *                                                        |
|          |          |          |          |          | 420 * NOTE: start data on an address that is easy to display |
|          |          |          |          |          | 421 * within Hercules                                        |
|          |          |          |          |          | 422 *                                                        |
|          |          |          |          |          | 423 *=====                                                   |
|          |          |          |          |          | 424                                                          |
| 00000528 |          |          | 00000528 | 00001000 | 425 ORG CFFTST+X'1000'                                       |
| 00001000 | 00000000 |          |          |          | 426 FAILED DC F'0' some test failed?                         |
| 00001004 | 00000000 |          |          |          | 427 TESTING DC F'0' current test number                      |
|          |          |          |          |          |                                                              |
|          |          |          |          |          | 430 *****                                                    |
|          |          |          |          |          | 431 * TEST failed : result messgae                           |
|          |          |          |          |          | 432 *****                                                    |
|          |          |          |          |          | 433 *                                                        |
|          |          |          |          |          | 434 * failed message and associated editting                 |
|          |          |          |          |          | 435 *                                                        |
| 00001008 | 40404040 | 40404040 |          |          | 436 PRTLNE DC C' Test # '                                    |
| 00001018 | A7A7A7   |          |          |          | 437 PRTNUM DC C'xxx'                                         |
| 0000101B | 40868189 | 93858440 |          |          | 438 DC C' failed for instruction '                           |
| 00001033 | A7A7A7A7 | A7A7A7A7 |          |          | 439 PRTNAME DC CL8'xxxxxxxxx'                                |
|          |          |          | 00000033 | 00000001 | 440 PRTLNG EQU *-PRTLNE                                      |
|          |          |          |          |          | 441                                                          |
|          |          |          |          |          | 442                                                          |
|          |          |          |          |          | 443 *****                                                    |
|          |          |          |          |          | 444 * TEST failed : CC message                               |
|          |          |          |          |          | 445 *****                                                    |
|          |          |          |          |          | 446 *                                                        |
|          |          |          |          |          | 447 * failed message and associated editting                 |
|          |          |          |          |          | 448 *                                                        |
| 0000103B | 40404040 | 40404040 |          |          | 449 CCPRTLNE DC C' Test # '                                  |
| 0000104B | A7A7A7   |          |          |          | 450 CCPRTNUM DC C'xxx'                                       |
| 0000104E | 40A69996 | 95874083 |          |          | 451 DC c' wrong cc for instruction '                         |
| 00001068 | A7A7A7A7 | A7A7A7A7 |          |          | 452 CCPRTNAME DC CL8'xxxxxxxxx'                              |
| 00001070 | 4085A797 | 8583A385 |          |          | 453 DC C' expected: cc='                                     |
| 0000107E | A7       |          |          |          | 454 CCPRTEXP DC C'x'                                         |
| 0000107F | 6B       |          |          |          | 455 DC C','                                                  |
| 00001080 | 40998583 | 8589A585 |          |          | 456 DC C' received: cc='                                     |
| 0000108E | A7       |          |          |          | 457 CCPRTGOT DC C'x'                                         |
| 0000108F | 4B       |          |          |          | 458 DC C'.'                                                  |
|          |          |          | 00000055 | 00000001 | 459 CCPRTLNG EQU *-CCPRTLNE                                  |









| LOC | OBJECT | CODE | ADDR1 | ADDR2 | STMT                                             |
|-----|--------|------|-------|-------|--------------------------------------------------|
|     |        |      |       |       | 532 *****                                        |
|     |        |      |       |       | 533 * (pending C8x7 CALG - Compare And Load Long |
|     |        |      |       |       | 534 * inclusion in SATK ASAM)                    |
|     |        |      |       |       | 535 *                                            |
|     |        |      |       |       | 536 * CALG Macro to help build CALG instruction  |
|     |        |      |       |       | 537 * Note: R1, R2, R3 are fixed registers:      |
|     |        |      |       |       | 538 * r1 = 1; r2 = 2; r3 = 3                     |
|     |        |      |       |       | 539 *****                                        |
|     |        |      |       |       | 540 MACRO                                        |
|     |        |      |       |       | 541 CALG                                         |
|     |        |      |       |       | 542 SPACE 1                                      |
|     |        |      |       |       | 543 DS 0H                                        |
|     |        |      |       |       | 544 DC XL6'C83710002000' CALG r3,0(R1),0(R2)     |
|     |        |      |       |       | 545 SPACE 1                                      |
|     |        |      |       |       | 546 MEND                                         |





| LOC | OBJECT CODE | ADDR1 | ADDR2 | STMT                                                                          |                                     |
|-----|-------------|-------|-------|-------------------------------------------------------------------------------|-------------------------------------|
|     |             |       |       | 564 *****                                                                     |                                     |
|     |             |       |       | 565 *       SSF Macro to help build test tables                               |                                     |
|     |             |       |       | 566 *****                                                                     |                                     |
|     |             |       |       | 567           MACRO                                                           |                                     |
|     |             |       |       | 568           SSF    &INST,&CC                                                |                                     |
|     |             |       |       | 569 .*                                                                        | &INST - instruction under test      |
|     |             |       |       | 570 .*                                                                        | &CC    - expected CC                |
|     |             |       |       | 571 .*                                                                        |                                     |
|     |             |       |       | 572           LCLA   &XCC(4)   &CC has mask values for FAILED condition codes |                                     |
|     |             |       |       | 573 &XCC(1)   SETA   7           CC != 0                                      |                                     |
|     |             |       |       | 574 &XCC(2)   SETA   11          CC != 1                                      |                                     |
|     |             |       |       | 575 &XCC(3)   SETA   13          CC != 2                                      |                                     |
|     |             |       |       | 576 &XCC(4)   SETA   14          CC != 3                                      |                                     |
|     |             |       |       | 577                                                                           |                                     |
|     |             |       |       | 578           GBLA   &TNUM                                                    |                                     |
|     |             |       |       | 579 &TNUM     SETA   &TNUM+1                                                  |                                     |
|     |             |       |       | 580                                                                           |                                     |
|     |             |       |       | 581           DS     0FD                                                      |                                     |
|     |             |       |       | 582           USING *,R5                                                      | base for test data and test routine |
|     |             |       |       | 583                                                                           |                                     |
|     |             |       |       | 584 T&TNUM    DC     A(X&TNUM)                                                | address of test routine             |
|     |             |       |       | 585           DC     H'&TNUM'                                                 | test number                         |
|     |             |       |       | 586           DC     XL1'00'                                                  |                                     |
|     |             |       |       | 587           DC     HL1'&CC'                                                 | cc                                  |
|     |             |       |       | 588           DC     HL1'&XCC(&CC+1)'                                         | cc failed mask                      |
|     |             |       |       | 589           DC     XL1'00'                                                  |                                     |
|     |             |       |       | 590                                                                           |                                     |
|     |             |       |       | 591           DC     CL8'&INST'                                               | instruction name                    |
|     |             |       |       | 592 01A&TNUM DC     AD(RE&TNUM+8)                                             | address of op1 source               |
|     |             |       |       | 593 02A&TNUM DC     AD(RE&TNUM+16)                                            | address of op2 source               |
|     |             |       |       | 594 03A&TNUM DC     AD(RE&TNUM+24)                                            | address of op3/R3 source            |
|     |             |       |       | 595           DC     A(8)                                                     | result length                       |
|     |             |       |       | 596 REA&TNUM DC     AD(RE&TNUM)                                               | result address                      |
|     |             |       |       | 597           DS     FD                                                       | gap                                 |
|     |             |       |       | 598 R30&TNUM DS     D                                                         | Op3/R3 Output                       |
|     |             |       |       | 599           DS     FD                                                       | gap                                 |
|     |             |       |       | 600 CCP&TNUM DS     2F                                                        | extract PSW after test (has CC)     |
|     |             |       |       | 601 CCF&TNUM DS     X                                                         | extracted cc                        |
|     |             |       |       | 602           DS     FD                                                       |                                     |
|     |             |       |       | 603 .*                                                                        |                                     |
|     |             |       |       | 604 *                                                                         | INSTRUCTION UNDER TEST ROUTINE      |
|     |             |       |       | 605 X&TNUM    DS     0F                                                       |                                     |
|     |             |       |       | 606           LG     R1,01A&TNUM                                              | get R1 source                       |
|     |             |       |       | 607           LG     R2,02A&TNUM                                              | get R2 source                       |
|     |             |       |       | 608           LG     R3,03A&TNUM                                              | get R3 source address               |
|     |             |       |       | 609           LG     R3,0(R3)                                                 | get R3 source value                 |
|     |             |       |       | 610                                                                           |                                     |
|     |             |       |       | 611           &INST                                                           | test instruction                    |
|     |             |       |       | 612                                                                           |                                     |
|     |             |       |       | 613           STG    R3,R30&TNUM                                              |                                     |
|     |             |       |       | 614                                                                           |                                     |
|     |             |       |       | 615           EPSW   R4,R0                                                    | extract psw                         |
|     |             |       |       | 616           ST     R4,CCP&TNUM                                              | to save CC                          |
|     |             |       |       | 617                                                                           |                                     |
|     |             |       |       | 618           BR     R11                                                      |                                     |
|     |             |       |       | 619                                                                           |                                     |



|     |        |                                                                 |                      |                 |
|-----|--------|-----------------------------------------------------------------|----------------------|-----------------|
| 625 | *****  |                                                                 |                      |                 |
| 626 | *      | PTTABLE Macro to generate table of pointers to individual tests |                      |                 |
| 627 | *****  |                                                                 |                      |                 |
| 628 |        |                                                                 |                      |                 |
| 629 |        | MACRO                                                           |                      |                 |
| 630 |        | PTTABLE                                                         |                      |                 |
| 631 |        | GBLA                                                            | &TNUM                |                 |
| 632 |        | LCLA                                                            | &CUR                 |                 |
| 633 | &CUR   | SETA                                                            | 1                    |                 |
| 634 | . *    |                                                                 |                      |                 |
| 635 | TTABLE | DS                                                              | 0F                   |                 |
| 636 | . LOOP | ANOP                                                            |                      |                 |
| 637 | . *    |                                                                 |                      |                 |
| 638 |        | DC                                                              | A(T&CUR)             | address of test |
| 639 | . *    |                                                                 |                      |                 |
| 640 | &CUR   | SETA                                                            | &CUR+1               |                 |
| 641 |        | AIF                                                             | (&CUR LE &TNUM).LOOP |                 |
| 642 | *      |                                                                 |                      |                 |
| 643 |        | DC                                                              | A(0)                 | END OF TABLE    |
| 644 |        | DC                                                              | A(0)                 |                 |
| 645 | . *    |                                                                 |                      |                 |
| 646 |        | MEND                                                            |                      |                 |

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|----------|-------------------|----------|----------|------------|---------------------------------------------|
|          |                   |          |          | 648 *****  |                                             |
|          |                   |          |          | 649 *      | Concurrent-functions facility tests         |
|          |                   |          |          | 650 *****  |                                             |
| 00001170 |                   | 00000000 | 00001F47 | 651 CFFTST | CSECT ,                                     |
|          |                   |          |          | 652 DS     | 0F                                          |
|          |                   |          |          | 654        | PRINT DATA                                  |
|          |                   |          |          | 655 *      |                                             |
|          |                   |          |          | 656 *      | C8x6 CAL - Compare And Load                 |
|          |                   |          |          | 657 *      | C8x7 CALG - Compare And Load Long           |
|          |                   |          |          | 658 *      | C8xF CALGF - Compare And Load Long Fullword |
|          |                   |          |          | 659 *      |                                             |
|          |                   |          |          | 660 *      | SSF instr, cc                               |
|          |                   |          |          | 661 *      | followed by:                                |
|          |                   |          |          | 662 *      | result - 8 bytes                            |
|          |                   |          |          | 663 *      | op1 - 8 byte source                         |
|          |                   |          |          | 664 *      | op2 - 8 byte source                         |
|          |                   |          |          | 665 *      | r3 - 8 byte source                          |
|          |                   |          |          | 666 *      | -----                                       |
|          |                   |          |          | 667 *      | C8x6 CAL - Compare And Load (32 bits)       |
|          |                   |          |          | 668 *      | -----                                       |
|          |                   |          |          | 669 *      | CC = 0 (equal) if r3 = op1; r3 = op2        |
| 00001170 |                   |          |          | 670        | SSF CAL,0                                   |
| 00001170 |                   | 00001170 |          | 671+       | DS 0FD                                      |
| 00001170 | 000011E0          |          |          | 672+       | USING *,R5                                  |
| 00001170 | 0001              |          |          | 673+T1     | DC A(X1)                                    |
| 00001174 | 0001              |          |          | 674+       | DC H'1'                                     |
| 00001176 | 00                |          |          | 675+       | DC XL1'00'                                  |
| 00001177 | 00                |          |          | 676+       | DC HL1'0'                                   |
| 00001178 | 07                |          |          | 677+       | DC HL1'7'                                   |
| 00001179 | 00                |          |          | 678+       | DC XL1'00'                                  |
| 0000117A | C3C1D340 40404040 |          |          | 679+       | DC CL8'CAL'                                 |
| 00001188 | 00000000 00001218 |          |          | 680+01A1   | DC AD(RE1+8)                                |
| 00001190 | 00000000 00001220 |          |          | 681+02A1   | DC AD(RE1+16)                               |
| 00001198 | 00000000 00001228 |          |          | 682+03A1   | DC AD(RE1+24)                               |
| 000011A0 | 00000008          |          |          | 683+       | DC A(8)                                     |
| 000011A8 | 00000000 00001210 |          |          | 684+REA1   | DC AD(RE1)                                  |
| 000011B0 | 00000000 00000000 |          |          | 685+       | DS FD                                       |
| 000011B8 | 00000000 00000000 |          |          | 686+R301   | DS D                                        |
| 000011C0 | 00000000 00000000 |          |          | 687+       | DS FD                                       |
| 000011C8 | 00000000 00000000 |          |          | 688+CCP1   | DS 2F                                       |
| 000011D0 | 00                |          |          | 689+CCF1   | DS X                                        |
| 000011D8 | 00000000 00000000 |          |          | 690+       | DS FD                                       |
|          |                   |          |          | 691+*      | INSTRUCTION UNDER TEST ROUTINE              |
| 000011E0 |                   |          |          | 692+X1     | DS 0F                                       |
| 000011E0 | E310 5018 0004    |          | 00001188 | 693+       | LG R1,01A1                                  |
| 000011E6 | E320 5020 0004    |          | 00001190 | 694+       | LG R2,02A1                                  |
| 000011EC | E330 5028 0004    |          | 00001198 | 695+       | LG R3,03A1                                  |
| 000011F2 | E333 0000 0004    |          | 00000000 | 696+       | LG R3,0(R3)                                 |
|          |                   |          |          |            | get R1 source                               |
|          |                   |          |          |            | get R2 source                               |
|          |                   |          |          |            | get R3 source address                       |
|          |                   |          |          |            | get R3 source value                         |
| 000011F8 |                   |          |          | 699+       | DS 0H                                       |
| 000011F8 | C8361000 2000     |          |          | 700+       | DC XL6'C83610002000'                        |
|          |                   |          |          |            | CAL r3,0(R1),0(R2)                          |
| 000011FE | E330 5048 0024    |          | 000011B8 | 702+       | STG R3,R301                                 |
| 00001204 | B98D 0040         |          |          | 703+       | EPSW R4,R0                                  |
|          |                   |          |          |            | extract psw                                 |

| LOC      | OBJECT CODE        | ADDR1    | ADDR2    | STMT     |       |                         |                                     |
|----------|--------------------|----------|----------|----------|-------|-------------------------|-------------------------------------|
| 00001208 | 5040 5058          |          | 000011C8 | 704+     | ST    | R4,CCP1                 | to save CC                          |
| 0000120C | 07FB               |          |          | 705+     | BR    | R11                     |                                     |
| 00001210 |                    |          |          | 706+RE1  | DC    | 0F                      |                                     |
| 00001210 |                    |          |          | 707+     | DROP  | R5                      |                                     |
| 00001210 | FF000000 00000004  |          |          | 708      | DC    | XL8'FF000000 00000004'  | result                              |
| 00001218 | 00000001 000000AA  |          |          | 709      | DC    | XL8'00000001 000000AA'  | op1 (first 32 bits)                 |
| 00001220 | 00000004 000000BB  |          |          | 710      | DC    | XL8'00000004 000000BB'  | op2 (first 32 bits)                 |
| 00001228 | FF000000 00000001  |          |          | 711      | DC    | XL8'FF000000 00000001'  | r3                                  |
|          |                    |          |          | 712      |       |                         |                                     |
|          |                    |          |          | 713      | SSF   | CAL,0                   |                                     |
| 00001230 |                    |          |          | 714+     | DS    | 0FD                     |                                     |
| 00001230 |                    | 00001230 |          | 715+     | USING | *,R5                    | base for test data and test routine |
| 00001230 | 000012A0           |          |          | 716+T2   | DC    | A(X2)                   | address of test routine             |
| 00001234 | 0002               |          |          | 717+     | DC    | H'2'                    | test number                         |
| 00001236 | 00                 |          |          | 718+     | DC    | XL1'00'                 |                                     |
| 00001237 | 00                 |          |          | 719+     | DC    | HL1'0'                  | cc                                  |
| 00001238 | 07                 |          |          | 720+     | DC    | HL1'7'                  | cc failed mask                      |
| 00001239 | 00                 |          |          | 721+     | DC    | XL1'00'                 |                                     |
| 0000123A | C3C1D340 40404040  |          |          | 722+     | DC    | CL8'CAL'                | instruction name                    |
| 00001248 | 00000000 000012D8  |          |          | 723+01A2 | DC    | AD(RE2+8)               | address of op1 source               |
| 00001250 | 00000000 000012E0  |          |          | 724+02A2 | DC    | AD(RE2+16)              | address of op2 source               |
| 00001258 | 00000000 000012E8  |          |          | 725+03A2 | DC    | AD(RE2+24)              | address of op3/R3 source            |
| 00001260 | 00000008           |          |          | 726+     | DC    | A(8)                    | result length                       |
| 00001268 | 00000000 000012D0  |          |          | 727+REA2 | DC    | AD(RE2)                 | result address                      |
| 00001270 | 00000000 00000000  |          |          | 728+     | DS    | FD                      | gap                                 |
| 00001278 | 00000000 00000000  |          |          | 729+R302 | DS    | D                       | Op3/R3 Output                       |
| 00001280 | 00000000 00000000  |          |          | 730+     | DS    | FD                      | gap                                 |
| 00001288 | 00000000 00000000  |          |          | 731+CCP2 | DS    | 2F                      | extract PSW after test (has CC)     |
| 00001290 | 00                 |          |          | 732+CCF2 | DS    | X                       | extracted cc                        |
| 00001298 | 00000000 00000000  |          |          | 733+     | DS    | FD                      |                                     |
|          |                    |          |          | 734+*    |       |                         | INSTRUCTION UNDER TEST ROUTINE      |
| 000012A0 |                    |          |          | 735+X2   | DS    | 0F                      |                                     |
| 000012A0 | E310 5018 0004     |          | 00001248 | 736+     | LG    | R1,01A2                 | get R1 source                       |
| 000012A6 | E320 5020 0004     |          | 00001250 | 737+     | LG    | R2,02A2                 | get R2 source                       |
| 000012AC | E330 5028 0004     |          | 00001258 | 738+     | LG    | R3,03A2                 | get R3 source address               |
| 000012B2 | E333 0000 0004     |          | 00000000 | 739+     | LG    | R3,0(R3)                | get R3 source value                 |
|          |                    |          |          | 742+     | DS    | 0H                      |                                     |
| 000012B8 | C8361000 2000      |          |          | 743+     | DC    | XL6'C83610002000'       | CAL r3,0(R1),0(R2)                  |
|          |                    |          |          | 745+     | STG   | R3,R302                 |                                     |
| 000012BE | E330 5048 0024     |          | 00001278 | 746+     | EPSW  | R4,R0                   | extract psw                         |
| 000012C4 | B98D 0040          |          |          | 747+     | ST    | R4,CCP2                 | to save CC                          |
| 000012C8 | 5040 5058          |          | 00001288 | 748+     | BR    | R11                     |                                     |
| 000012CC | 07FB               |          |          | 749+RE2  | DC    | 0F                      |                                     |
| 000012D0 |                    |          |          | 750+     | DROP  | R5                      |                                     |
| 000012D0 | FFFFFFFF A0000004  |          |          | 751      | DC    | XL8'FFFFFFFF A0000004'  | result                              |
| 000012D8 | 00000001 AAAAAAAA  |          |          | 752      | DC    | XL8'00000001 AAAAAAAA'  | op1 (first 32 bits)                 |
| 000012E0 | A0000004 BBBB BBBB |          |          | 753      | DC    | XL8'A0000004 BBBB BBBB' | op2 (first 32 bits)                 |
| 000012E8 | FFFFFFFF 00000001  |          |          | 754      | DC    | XL8'FFFFFFFF 00000001'  | r3                                  |
|          |                    |          |          | 755      |       |                         |                                     |
|          |                    |          |          | 756      | SSF   | CAL,0                   |                                     |
| 000012F0 |                    |          |          | 757+     | DS    | 0FD                     |                                     |
| 000012F0 |                    | 000012F0 |          | 758+     | USING | *,R5                    | base for test data and test routine |
| 000012F0 | 00001360           |          |          | 759+T3   | DC    | A(X3)                   | address of test routine             |
| 000012F4 | 0003               |          |          | 760+     | DC    | H'3'                    | test number                         |

| LOC      | OBJECT CODE        | ADDR1    | ADDR2    | STMT     |                                             |                         |                                     |
|----------|--------------------|----------|----------|----------|---------------------------------------------|-------------------------|-------------------------------------|
| 000012F6 | 00                 |          |          | 761+     | DC                                          | XL1'00'                 |                                     |
| 000012F7 | 00                 |          |          | 762+     | DC                                          | HL1'0'                  | cc                                  |
| 000012F8 | 07                 |          |          | 763+     | DC                                          | HL1'7'                  | cc failed mask                      |
| 000012F9 | 00                 |          |          | 764+     | DC                                          | XL1'00'                 |                                     |
| 000012FA | C3C1D340 40404040  |          |          | 765+     | DC                                          | CL8'CAL'                | instruction name                    |
| 00001308 | 00000000 00001398  |          |          | 766+01A3 | DC                                          | AD(RE3+8)               | address of op1 source               |
| 00001310 | 00000000 000013A0  |          |          | 767+02A3 | DC                                          | AD(RE3+16)              | address of op2 source               |
| 00001318 | 00000000 000013A8  |          |          | 768+03A3 | DC                                          | AD(RE3+24)              | address of op3/R3 source            |
| 00001320 | 00000008           |          |          | 769+     | DC                                          | A(8)                    | result length                       |
| 00001328 | 00000000 00001390  |          |          | 770+REA3 | DC                                          | AD(RE3)                 | result address                      |
| 00001330 | 00000000 00000000  |          |          | 771+     | DS                                          | FD                      | gap                                 |
| 00001338 | 00000000 00000000  |          |          | 772+R303 | DS                                          | D                       | Op3/R3 Output                       |
| 00001340 | 00000000 00000000  |          |          | 773+     | DS                                          | FD                      | gap                                 |
| 00001348 | 00000000 00000000  |          |          | 774+CCP3 | DS                                          | 2F                      | extract PSW after test (has CC)     |
| 00001350 | 00                 |          |          | 775+CCF3 | DS                                          | X                       | extracted cc                        |
| 00001358 | 00000000 00000000  |          |          | 776+     | DS                                          | FD                      |                                     |
|          |                    |          |          | 777+*    |                                             |                         | INSTRUCTION UNDER TEST ROUTINE      |
| 00001360 |                    |          |          | 778+X3   | DS                                          | 0F                      |                                     |
| 00001360 | E310 5018 0004     |          | 00001308 | 779+     | LG                                          | R1,01A3                 | get R1 source                       |
| 00001366 | E320 5020 0004     |          | 00001310 | 780+     | LG                                          | R2,02A3                 | get R2 source                       |
| 0000136C | E330 5028 0004     |          | 00001318 | 781+     | LG                                          | R3,03A3                 | get R3 source address               |
| 00001372 | E333 0000 0004     |          | 00000000 | 782+     | LG                                          | R3,0(R3)                | get R3 source value                 |
| 00001378 |                    |          |          | 785+     | DS                                          | 0H                      |                                     |
| 00001378 | C8361000 2000      |          |          | 786+     | DC                                          | XL6'C83610002000'       | CAL r3,0(R1),0(R2)                  |
| 0000137E | E330 5048 0024     |          | 00001338 | 788+     | STG                                         | R3,R303                 |                                     |
| 00001384 | B98D 0040          |          |          | 789+     | EPSW                                        | R4,R0                   | extract psw                         |
| 00001388 | 5040 5058          |          | 00001348 | 790+     | ST                                          | R4,CCP3                 | to save CC                          |
| 0000138C | 07FB               |          |          | 791+     | BR                                          | R11                     |                                     |
| 00001390 |                    |          |          | 792+RE3  | DC                                          | 0F                      |                                     |
| 00001390 |                    |          |          | 793+     | DROP                                        | R5                      |                                     |
| 00001390 | FFFFFFFF 99999999  |          |          | 794      | DC                                          | XL8'FFFFFFFF 99999999'  | result                              |
| 00001398 | 00000001 AAAAAAAA  |          |          | 795      | DC                                          | XL8'00000001 AAAAAAAA'  | op1 (first 32 bits)                 |
| 000013A0 | 99999999 BBBB BBBB |          |          | 796      | DC                                          | XL8'99999999 BBBB BBBB' | op2 (first 32 bits)                 |
| 000013A8 | FFFFFFFF 00000001  |          |          | 797      | DC                                          | XL8'FFFFFFFF 00000001'  | r3                                  |
|          |                    |          |          | 798      |                                             |                         |                                     |
|          |                    |          |          | 799      | * CC = 1 (not equal) if r3 != op1; r3 = op1 |                         |                                     |
| 000013B0 |                    |          |          | 800      | SSF                                         | CAL,1                   |                                     |
| 000013B0 |                    | 000013B0 |          | 801+     | DS                                          | 0FD                     |                                     |
| 000013B0 | 00001420           |          |          | 802+     | USING                                       | *,R5                    | base for test data and test routine |
| 000013B4 | 0004               |          |          | 803+T4   | DC                                          | A(X4)                   | address of test routine             |
| 000013B6 | 00                 |          |          | 804+     | DC                                          | H'4'                    | test number                         |
| 000013B7 | 01                 |          |          | 805+     | DC                                          | XL1'00'                 |                                     |
| 000013B8 | 0B                 |          |          | 806+     | DC                                          | HL1'1'                  | cc                                  |
| 000013B9 | 00                 |          |          | 807+     | DC                                          | HL1'11'                 | cc failed mask                      |
| 000013BA | C3C1D340 40404040  |          |          | 808+     | DC                                          | XL1'00'                 |                                     |
| 000013C8 | 00000000 00001458  |          |          | 809+     | DC                                          | CL8'CAL'                | instruction name                    |
| 000013D0 | 00000000 00001460  |          |          | 810+01A4 | DC                                          | AD(RE4+8)               | address of op1 source               |
| 000013D8 | 00000000 00001468  |          |          | 811+02A4 | DC                                          | AD(RE4+16)              | address of op2 source               |
| 000013E0 | 00000008           |          |          | 812+03A4 | DC                                          | AD(RE4+24)              | address of op3/R3 source            |
| 000013E8 | 00000000 00001450  |          |          | 813+     | DC                                          | A(8)                    | result length                       |
| 000013F0 | 00000000 00000000  |          |          | 814+REA4 | DC                                          | AD(RE4)                 | result address                      |
| 000013F8 | 00000000 00000000  |          |          | 815+     | DS                                          | FD                      | gap                                 |
| 00001400 | 00000000 00000000  |          |          | 816+R304 | DS                                          | D                       | Op3/R3 Output                       |
|          |                    |          |          | 817+     | DS                                          | FD                      | gap                                 |



| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT     |       |                        |                                     |
|----------|-------------------|----------|----------|----------|-------|------------------------|-------------------------------------|
| 00001408 | 00000000 00000000 |          |          | 818+CCP4 | DS    | 2F                     | extract PSW after test (has CC)     |
| 00001410 | 00                |          |          | 819+CCF4 | DS    | X                      | extracted cc                        |
| 00001418 | 00000000 00000000 |          |          | 820+     | DS    | FD                     |                                     |
|          |                   |          |          | 821+*    |       |                        | INSTRUCTION UNDER TEST ROUTINE      |
| 00001420 |                   |          |          | 822+X4   | DS    | 0F                     |                                     |
| 00001420 | E310 5018 0004    |          | 000013C8 | 823+     | LG    | R1,01A4                | get R1 source                       |
| 00001426 | E320 5020 0004    |          | 000013D0 | 824+     | LG    | R2,02A4                | get R2 source                       |
| 0000142C | E330 5028 0004    |          | 000013D8 | 825+     | LG    | R3,03A4                | get R3 source address               |
| 00001432 | E333 0000 0004    |          | 00000000 | 826+     | LG    | R3,0(R3)               | get R3 source value                 |
| 00001438 |                   |          |          | 829+     | DS    | 0H                     |                                     |
| 00001438 | C8361000 2000     |          |          | 830+     | DC    | XL6'C83610002000'      | CAL r3,0(R1),0(R2)                  |
| 0000143E | E330 5048 0024    |          | 000013F8 | 832+     | STG   | R3,R304                |                                     |
| 00001444 | B98D 0040         |          |          | 833+     | EPSW  | R4,R0                  | extract psw                         |
| 00001448 | 5040 5058         |          | 00001408 | 834+     | ST    | R4,CCP4                | to save CC                          |
| 0000144C | 07FB              |          |          | 835+     | BR    | R11                    |                                     |
| 00001450 |                   |          |          | 836+RE4  | DC    | 0F                     |                                     |
| 00001450 |                   |          |          | 837+     | DROP  | R5                     |                                     |
| 00001450 | FF000000 00000001 |          |          | 838      | DC    | XL8'FF000000 00000001' | result                              |
| 00001458 | 00000001 000000FF |          |          | 839      | DC    | XL8'00000001 000000FF' | op1 (first 32 bits)                 |
| 00001460 | 00000004 000000FF |          |          | 840      | DC    | XL8'00000004 000000FF' | op2 (first 32 bits)                 |
| 00001468 | FF000000 00000000 |          |          | 841      | DC    | XL8'FF000000 00000000' | r3                                  |
|          |                   |          |          | 842      |       |                        |                                     |
| 00001470 |                   |          |          | 843      | SSF   | CAL,1                  |                                     |
| 00001470 |                   | 00001470 |          | 844+     | DS    | 0FD                    |                                     |
| 00001470 | 000014E0          |          |          | 845+     | USING | *,R5                   | base for test data and test routine |
| 00001474 | 0005              |          |          | 846+T5   | DC    | A(X5)                  | address of test routine             |
| 00001474 | 00                |          |          | 847+     | DC    | H'5'                   | test number                         |
| 00001476 | 00                |          |          | 848+     | DC    | XL1'00'                |                                     |
| 00001477 | 01                |          |          | 849+     | DC    | HL1'1'                 | cc                                  |
| 00001478 | 0B                |          |          | 850+     | DC    | HL1'11'                | cc failed mask                      |
| 00001479 | 00                |          |          | 851+     | DC    | XL1'00'                |                                     |
| 0000147A | C3C1D340 40404040 |          |          | 852+     | DC    | CL8'CAL'               | instruction name                    |
| 00001488 | 00000000 00001518 |          |          | 853+01A5 | DC    | AD(RE5+8)              | address of op1 source               |
| 00001490 | 00000000 00001520 |          |          | 854+02A5 | DC    | AD(RE5+16)             | address of op2 source               |
| 00001498 | 00000000 00001528 |          |          | 855+03A5 | DC    | AD(RE5+24)             | address of op3/R3 source            |
| 000014A0 | 00000008          |          |          | 856+     | DC    | A(8)                   | result length                       |
| 000014A8 | 00000000 00001510 |          |          | 857+REA5 | DC    | AD(RE5)                | result address                      |
| 000014B0 | 00000000 00000000 |          |          | 858+     | DS    | FD                     | gap                                 |
| 000014B8 | 00000000 00000000 |          |          | 859+R305 | DS    | D                      | Op3/R3 Output                       |
| 000014C0 | 00000000 00000000 |          |          | 860+     | DS    | FD                     | gap                                 |
| 000014C8 | 00000000 00000000 |          |          | 861+CCP5 | DS    | 2F                     | extract PSW after test (has CC)     |
| 000014D0 | 00                |          |          | 862+CCF5 | DS    | X                      | extracted cc                        |
| 000014D8 | 00000000 00000000 |          |          | 863+     | DS    | FD                     |                                     |
|          |                   |          |          | 864+*    |       |                        | INSTRUCTION UNDER TEST ROUTINE      |
| 000014E0 |                   |          |          | 865+X5   | DS    | 0F                     |                                     |
| 000014E0 | E310 5018 0004    |          | 00001488 | 866+     | LG    | R1,01A5                | get R1 source                       |
| 000014E6 | E320 5020 0004    |          | 00001490 | 867+     | LG    | R2,02A5                | get R2 source                       |
| 000014EC | E330 5028 0004    |          | 00001498 | 868+     | LG    | R3,03A5                | get R3 source address               |
| 000014F2 | E333 0000 0004    |          | 00000000 | 869+     | LG    | R3,0(R3)               | get R3 source value                 |
| 000014F8 |                   |          |          | 872+     | DS    | 0H                     |                                     |
| 000014F8 | C8361000 2000     |          |          | 873+     | DC    | XL6'C83610002000'      | CAL r3,0(R1),0(R2)                  |
| 000014FE | E330 5048 0024    |          | 000014B8 | 875+     | STG   | R3,R305                |                                     |



| LOC      | OBJECT CODE        | ADDR1    | ADDR2    | STMT     |                                               |                         |                                     |
|----------|--------------------|----------|----------|----------|-----------------------------------------------|-------------------------|-------------------------------------|
| 00001504 | B98D 0040          |          |          | 876+     | EPSW                                          | R4,R0                   | extract psw                         |
| 00001508 | 5040 5058          |          | 000014C8 | 877+     | ST                                            | R4,CCP5                 | to save CC                          |
| 0000150C | 07FB               |          |          | 878+     | BR                                            | R11                     |                                     |
| 00001510 |                    |          |          | 879+RE5  | DC                                            | 0F                      |                                     |
| 00001510 |                    |          |          | 880+     | DROP                                          | R5                      |                                     |
| 00001510 | FFFFFFFF 00000DDD  |          |          | 881      | DC                                            | XL8'FFFFFFFF 00000DDD'  | result                              |
| 00001518 | 00000DDD AAAAAAAA  |          |          | 882      | DC                                            | XL8'00000DDD AAAAAAAA'  | op1 (first 32 bits)                 |
| 00001520 | A0000004 BBBB BBBB |          |          | 883      | DC                                            | XL8'A0000004 BBBB BBBB' | op2 (first 32 bits)                 |
| 00001528 | FFFFFFFF 0000CCCC  |          |          | 884      | DC                                            | XL8'FFFFFFFF 0000CCCC'  | r3                                  |
|          |                    |          |          | 885      |                                               |                         |                                     |
|          |                    |          |          | 886      | SSF                                           | CAL,1                   |                                     |
| 00001530 |                    |          |          | 887+     | DS                                            | 0FD                     |                                     |
| 00001530 |                    | 00001530 |          | 888+     | USING                                         | *,R5                    | base for test data and test routine |
| 00001530 | 000015A0           |          |          | 889+T6   | DC                                            | A(X6)                   | address of test routine             |
| 00001534 | 0006               |          |          | 890+     | DC                                            | H'6'                    | test number                         |
| 00001536 | 00                 |          |          | 891+     | DC                                            | XL1'00'                 |                                     |
| 00001537 | 01                 |          |          | 892+     | DC                                            | HL1'1'                  | cc                                  |
| 00001538 | 0B                 |          |          | 893+     | DC                                            | HL1'11'                 | cc failed mask                      |
| 00001539 | 00                 |          |          | 894+     | DC                                            | XL1'00'                 |                                     |
| 0000153A | C3C1D340 40404040  |          |          | 895+     | DC                                            | CL8'CAL'                | instruction name                    |
| 00001548 | 00000000 000015D8  |          |          | 896+01A6 | DC                                            | AD(RE6+8)               | address of op1 source               |
| 00001550 | 00000000 000015E0  |          |          | 897+02A6 | DC                                            | AD(RE6+16)              | address of op2 source               |
| 00001558 | 00000000 000015E8  |          |          | 898+03A6 | DC                                            | AD(RE6+24)              | address of op3/R3 source            |
| 00001560 | 00000008           |          |          | 899+     | DC                                            | A(8)                    | result length                       |
| 00001568 | 00000000 000015D0  |          |          | 900+REA6 | DC                                            | AD(RE6)                 | result address                      |
| 00001570 | 00000000 00000000  |          |          | 901+     | DS                                            | FD                      | gap                                 |
| 00001578 | 00000000 00000000  |          |          | 902+R306 | DS                                            | D                       | Op3/R3 Output                       |
| 00001580 | 00000000 00000000  |          |          | 903+     | DS                                            | FD                      | gap                                 |
| 00001588 | 00000000 00000000  |          |          | 904+CCP6 | DS                                            | 2F                      | extract PSW after test (has CC)     |
| 00001590 | 00                 |          |          | 905+CCF6 | DS                                            | X                       | extracted cc                        |
| 00001598 | 00000000 00000000  |          |          | 906+     | DS                                            | FD                      |                                     |
|          |                    |          |          | 907+*    |                                               |                         | INSTRUCTION UNDER TEST ROUTINE      |
| 000015A0 |                    |          |          | 908+X6   | DS                                            | 0F                      |                                     |
| 000015A0 | E310 5018 0004     |          | 00001548 | 909+     | LG                                            | R1,01A6                 | get R1 source                       |
| 000015A6 | E320 5020 0004     |          | 00001550 | 910+     | LG                                            | R2,02A6                 | get R2 source                       |
| 000015AC | E330 5028 0004     |          | 00001558 | 911+     | LG                                            | R3,03A6                 | get R3 source address               |
| 000015B2 | E333 0000 0004     |          | 00000000 | 912+     | LG                                            | R3,0(R3)                | get R3 source value                 |
|          |                    |          |          |          |                                               |                         |                                     |
| 000015B8 |                    |          |          | 915+     | DS                                            | 0H                      |                                     |
| 000015B8 | C8361000 2000      |          |          | 916+     | DC                                            | XL6'C83610002000'       | CAL r3,0(R1),0(R2)                  |
|          |                    |          |          |          |                                               |                         |                                     |
| 000015BE | E330 5048 0024     |          | 00001578 | 918+     | STG                                           | R3,R306                 |                                     |
| 000015C4 | B98D 0040          |          |          | 919+     | EPSW                                          | R4,R0                   | extract psw                         |
| 000015C8 | 5040 5058          |          | 00001588 | 920+     | ST                                            | R4,CCP6                 | to save CC                          |
| 000015CC | 07FB               |          |          | 921+     | BR                                            | R11                     |                                     |
| 000015D0 |                    |          |          | 922+RE6  | DC                                            | 0F                      |                                     |
| 000015D0 |                    |          |          | 923+     | DROP                                          | R5                      |                                     |
| 000015D0 | FFFFFFFF 0000CCCC  |          |          | 924      | DC                                            | XL8'FFFFFFFF 0000CCCC'  | result                              |
| 000015D8 | 0000CCCC AAAAAAAA  |          |          | 925      | DC                                            | XL8'0000CCCC AAAAAAAA'  | op1 (first 32 bits)                 |
| 000015E0 | 99999999 BBBB BBBB |          |          | 926      | DC                                            | XL8'99999999 BBBB BBBB' | op2 (first 32 bits)                 |
| 000015E8 | FFFFFFFF 00009999  |          |          | 927      | DC                                            | XL8'FFFFFFFF 00009999'  | r3                                  |
|          |                    |          |          | 928      |                                               |                         |                                     |
|          |                    |          |          | 929      | *-----                                        |                         |                                     |
|          |                    |          |          | 930      | * C8x7 CALG - Compare And Load Long (64 bits) |                         |                                     |
|          |                    |          |          | 931      | *-----                                        |                         |                                     |
|          |                    |          |          | 932      | * CC = 0 (equal) if r3 = op1; r3 = op2        |                         |                                     |

| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT     |       |                        |
|----------|-------------------|----------|----------|----------|-------|------------------------|
| 000015F0 |                   |          |          | 933      | SSF   | CALG,0                 |
| 000015F0 |                   |          |          | 934+     | DS    | 0FD                    |
| 000015F0 |                   | 000015F0 |          | 935+     | USING | *,R5                   |
| 000015F0 | 00001660          |          |          | 936+T7   | DC    | A(X7)                  |
| 000015F4 | 0007              |          |          | 937+     | DC    | H'7'                   |
| 000015F6 | 00                |          |          | 938+     | DC    | XL1'00'                |
| 000015F7 | 00                |          |          | 939+     | DC    | HL1'0'                 |
| 000015F8 | 07                |          |          | 940+     | DC    | HL1'7'                 |
| 000015F9 | 00                |          |          | 941+     | DC    | XL1'00'                |
| 000015FA | C3C1D3C7 40404040 |          |          | 942+     | DC    | CL8'CALG'              |
| 00001608 | 00000000 00001698 |          |          | 943+01A7 | DC    | AD(RE7+8)              |
| 00001610 | 00000000 000016A0 |          |          | 944+02A7 | DC    | AD(RE7+16)             |
| 00001618 | 00000000 000016A8 |          |          | 945+03A7 | DC    | AD(RE7+24)             |
| 00001620 | 00000008          |          |          | 946+     | DC    | A(8)                   |
| 00001628 | 00000000 00001690 |          |          | 947+REA7 | DC    | AD(RE7)                |
| 00001630 | 00000000 00000000 |          |          | 948+     | DS    | FD                     |
| 00001638 | 00000000 00000000 |          |          | 949+R307 | DS    | D                      |
| 00001640 | 00000000 00000000 |          |          | 950+     | DS    | FD                     |
| 00001648 | 00000000 00000000 |          |          | 951+CCP7 | DS    | 2F                     |
| 00001650 | 00                |          |          | 952+CCF7 | DS    | X                      |
| 00001658 | 00000000 00000000 |          |          | 953+     | DS    | FD                     |
|          |                   |          |          | 954+*    |       |                        |
| 00001660 |                   |          |          | 955+X7   | DS    | 0F                     |
| 00001660 | E310 5018 0004    |          | 00001608 | 956+     | LG    | R1,01A7                |
| 00001666 | E320 5020 0004    |          | 00001610 | 957+     | LG    | R2,02A7                |
| 0000166C | E330 5028 0004    |          | 00001618 | 958+     | LG    | R3,03A7                |
| 00001672 | E333 0000 0004    |          | 00000000 | 959+     | LG    | R3,0(R3)               |
|          |                   |          |          |          |       |                        |
| 00001678 |                   |          |          | 962+     | DS    | 0H                     |
| 00001678 | C8371000 2000     |          |          | 963+     | DC    | XL6'C83710002000'      |
|          |                   |          |          |          |       | CALG r3,0(R1),0(R2)    |
| 0000167E | E330 5048 0024    |          | 00001638 | 965+     | STG   | R3,R307                |
| 00001684 | B98D 0040         |          |          | 966+     | EPSW  | R4,R0                  |
| 00001688 | 5040 5058         |          | 00001648 | 967+     | ST    | R4,CCP7                |
| 0000168C | 07FB              |          |          | 968+     | BR    | R11                    |
| 00001690 |                   |          |          | 969+RE7  | DC    | 0F                     |
| 00001690 |                   |          |          | 970+     | DROP  | R5                     |
| 00001690 | 00000004 000000BB |          |          | 971      | DC    | XL8'00000004 000000BB' |
| 00001698 | FF000000 00000001 |          |          | 972      | DC    | XL8'FF000000 00000001' |
| 000016A0 | 00000004 000000BB |          |          | 973      | DC    | XL8'00000004 000000BB' |
| 000016A8 | FF000000 00000001 |          |          | 974      | DC    | XL8'FF000000 00000001' |
|          |                   |          |          | 975      |       |                        |
| 000016B0 |                   |          |          | 976      | SSF   | CALG,0                 |
| 000016B0 |                   |          |          | 977+     | DS    | 0FD                    |
| 000016B0 |                   | 000016B0 |          | 978+     | USING | *,R5                   |
| 000016B0 | 00001720          |          |          | 979+T8   | DC    | A(X8)                  |
| 000016B4 | 0008              |          |          | 980+     | DC    | H'8'                   |
| 000016B6 | 00                |          |          | 981+     | DC    | XL1'00'                |
| 000016B7 | 00                |          |          | 982+     | DC    | HL1'0'                 |
| 000016B8 | 07                |          |          | 983+     | DC    | HL1'7'                 |
| 000016B9 | 00                |          |          | 984+     | DC    | XL1'00'                |
| 000016BA | C3C1D3C7 40404040 |          |          | 985+     | DC    | CL8'CALG'              |
| 000016C8 | 00000000 00001758 |          |          | 986+01A8 | DC    | AD(RE8+8)              |
| 000016D0 | 00000000 00001760 |          |          | 987+02A8 | DC    | AD(RE8+16)             |
| 000016D8 | 00000000 00001768 |          |          | 988+03A8 | DC    | AD(RE8+24)             |
| 000016E0 | 00000008          |          |          | 989+     | DC    | A(8)                   |

| LOC      | OBJECT CODE        | ADDR1    | ADDR2    | STMT      |       |                         |                                     |
|----------|--------------------|----------|----------|-----------|-------|-------------------------|-------------------------------------|
| 000016E8 | 00000000 00001750  |          |          | 990+REA8  | DC    | AD(RE8)                 | result address                      |
| 000016F0 | 00000000 00000000  |          |          | 991+      | DS    | FD                      | gap                                 |
| 000016F8 | 00000000 00000000  |          |          | 992+R308  | DS    | D                       | Op3/R3 Output                       |
| 00001700 | 00000000 00000000  |          |          | 993+      | DS    | FD                      | gap                                 |
| 00001708 | 00000000 00000000  |          |          | 994+CCP8  | DS    | 2F                      | extract PSW after test (has CC)     |
| 00001710 | 00                 |          |          | 995+CCF8  | DS    | X                       | extracted cc                        |
| 00001718 | 00000000 00000000  |          |          | 996+      | DS    | FD                      |                                     |
|          |                    |          |          | 997+*     |       |                         | INSTRUCTION UNDER TEST ROUTINE      |
| 00001720 |                    |          |          | 998+X8    | DS    | 0F                      |                                     |
| 00001720 | E310 5018 0004     |          | 000016C8 | 999+      | LG    | R1,01A8                 | get R1 source                       |
| 00001726 | E320 5020 0004     |          | 000016D0 | 1000+     | LG    | R2,02A8                 | get R2 source                       |
| 0000172C | E330 5028 0004     |          | 000016D8 | 1001+     | LG    | R3,03A8                 | get R3 source address               |
| 00001732 | E333 0000 0004     |          | 00000000 | 1002+     | LG    | R3,0(R3)                | get R3 source value                 |
| 00001738 |                    |          |          | 1005+     | DS    | 0H                      |                                     |
| 00001738 | C8371000 2000      |          |          | 1006+     | DC    | XL6'C83710002000'       | CALG r3,0(R1),0(R2)                 |
| 0000173E | E330 5048 0024     |          | 000016F8 | 1008+     | STG   | R3,R308                 |                                     |
| 00001744 | B98D 0040          |          |          | 1009+     | EPSW  | R4,R0                   | extract psw                         |
| 00001748 | 5040 5058          |          | 00001708 | 1010+     | ST    | R4,CCP8                 | to save CC                          |
| 0000174C | 07FB               |          |          | 1011+     | BR    | R11                     |                                     |
| 00001750 |                    |          |          | 1012+RE8  | DC    | 0F                      |                                     |
| 00001750 |                    |          |          | 1013+     | DROP  | R5                      |                                     |
| 00001750 | A0000004 BBBB BBBB |          |          | 1014      | DC    | XL8'A0000004 BBBB BBBB' | result                              |
| 00001758 | FFFFFFFF 00000001  |          |          | 1015      | DC    | XL8'FFFFFFFF 00000001'  | op1                                 |
| 00001760 | A0000004 BBBB BBBB |          |          | 1016      | DC    | XL8'A0000004 BBBB BBBB' | op2                                 |
| 00001768 | FFFFFFFF 00000001  |          |          | 1017      | DC    | XL8'FFFFFFFF 00000001'  | r3                                  |
|          |                    |          |          | 1018      |       |                         |                                     |
|          |                    |          |          | 1019      | SSF   | CALG,0                  |                                     |
| 00001770 |                    |          |          | 1020+     | DS    | 0FD                     |                                     |
| 00001770 |                    | 00001770 |          | 1021+     | USING | *,R5                    | base for test data and test routine |
| 00001770 | 000017E0           |          |          | 1022+T9   | DC    | A(X9)                   | address of test routine             |
| 00001774 | 0009               |          |          | 1023+     | DC    | H'9'                    | test number                         |
| 00001776 | 00                 |          |          | 1024+     | DC    | XL1'00'                 |                                     |
| 00001777 | 00                 |          |          | 1025+     | DC    | HL1'0'                  | cc                                  |
| 00001778 | 07                 |          |          | 1026+     | DC    | HL1'7'                  | cc failed mask                      |
| 00001779 | 00                 |          |          | 1027+     | DC    | XL1'00'                 |                                     |
| 0000177A | C3C1D3C7 40404040  |          |          | 1028+     | DC    | CL8'CALG'               | instruction name                    |
| 00001788 | 00000000 00001818  |          |          | 1029+01A9 | DC    | AD(RE9+8)               | address of op1 source               |
| 00001790 | 00000000 00001820  |          |          | 1030+02A9 | DC    | AD(RE9+16)              | address of op2 source               |
| 00001798 | 00000000 00001828  |          |          | 1031+03A9 | DC    | AD(RE9+24)              | address of op3/R3 source            |
| 000017A0 | 00000008           |          |          | 1032+     | DC    | A(8)                    | result length                       |
| 000017A8 | 00000000 00001810  |          |          | 1033+REA9 | DC    | AD(RE9)                 | result address                      |
| 000017B0 | 00000000 00000000  |          |          | 1034+     | DS    | FD                      | gap                                 |
| 000017B8 | 00000000 00000000  |          |          | 1035+R309 | DS    | D                       | Op3/R3 Output                       |
| 000017C0 | 00000000 00000000  |          |          | 1036+     | DS    | FD                      | gap                                 |
| 000017C8 | 00000000 00000000  |          |          | 1037+CCP9 | DS    | 2F                      | extract PSW after test (has CC)     |
| 000017D0 | 00                 |          |          | 1038+CCF9 | DS    | X                       | extracted cc                        |
| 000017D8 | 00000000 00000000  |          |          | 1039+     | DS    | FD                      |                                     |
|          |                    |          |          | 1040+*    |       |                         | INSTRUCTION UNDER TEST ROUTINE      |
| 000017E0 |                    |          |          | 1041+X9   | DS    | 0F                      |                                     |
| 000017E0 | E310 5018 0004     |          | 00001788 | 1042+     | LG    | R1,01A9                 | get R1 source                       |
| 000017E6 | E320 5020 0004     |          | 00001790 | 1043+     | LG    | R2,02A9                 | get R2 source                       |
| 000017EC | E330 5028 0004     |          | 00001798 | 1044+     | LG    | R3,03A9                 | get R3 source address               |
| 000017F2 | E333 0000 0004     |          | 00000000 | 1045+     | LG    | R3,0(R3)                | get R3 source value                 |

| LOC      | OBJECT CODE        | ADDR1    | ADDR2    | STMT       |                                             |                         |                                     |  |
|----------|--------------------|----------|----------|------------|---------------------------------------------|-------------------------|-------------------------------------|--|
| 000017F8 |                    |          |          | 1048+      | DS                                          | 0H                      |                                     |  |
| 000017F8 | C8371000 2000      |          |          | 1049+      | DC                                          | XL6'C83710002000'       | CALG r3,0(R1),0(R2)                 |  |
| 000017FE | E330 5048 0024     |          | 000017B8 | 1051+      | STG                                         | R3,R309                 |                                     |  |
| 00001804 | B98D 0040          |          |          | 1052+      | EPSW                                        | R4,R0                   | extract psw                         |  |
| 00001808 | 5040 5058          |          | 000017C8 | 1053+      | ST                                          | R4,CCP9                 | to save CC                          |  |
| 0000180C | 07FB               |          |          | 1054+      | BR                                          | R11                     |                                     |  |
| 00001810 |                    |          |          | 1055+RE9   | DC                                          | 0F                      |                                     |  |
| 00001810 |                    |          |          | 1056+      | DROP                                        | R5                      |                                     |  |
| 00001810 | 99999999 BBBB BBBB |          |          | 1057       | DC                                          | XL8'99999999 BBBB BBBB' | result                              |  |
| 00001818 | FFFFFF00 00000CCC  |          |          | 1058       | DC                                          | XL8'FFFFFF00 00000CCC'  | op1                                 |  |
| 00001820 | 99999999 BBBB BBBB |          |          | 1059       | DC                                          | XL8'99999999 BBBB BBBB' | op2                                 |  |
| 00001828 | FFFFFF00 00000CCC  |          |          | 1060       | DC                                          | XL8'FFFFFF00 00000CCC'  | r3                                  |  |
|          |                    |          |          | 1061       |                                             |                         |                                     |  |
|          |                    |          |          | 1062       | * CC = 1 (not equal) if r3 != op1; r3 = op1 |                         |                                     |  |
|          |                    |          |          | 1063       | SSF                                         | CALG,1                  |                                     |  |
| 00001830 |                    |          |          | 1064+      | DS                                          | 0FD                     |                                     |  |
| 00001830 |                    | 00001830 |          | 1065+      | USING                                       | *,R5                    | base for test data and test routine |  |
| 00001830 | 000018A0           |          |          | 1066+T10   | DC                                          | A(X10)                  | address of test routine             |  |
| 00001834 | 000A               |          |          | 1067+      | DC                                          | H'10'                   | test number                         |  |
| 00001836 | 00                 |          |          | 1068+      | DC                                          | XL1'00'                 |                                     |  |
| 00001837 | 01                 |          |          | 1069+      | DC                                          | HL1'1'                  | cc                                  |  |
| 00001838 | 0B                 |          |          | 1070+      | DC                                          | HL1'11'                 | cc failed mask                      |  |
| 00001839 | 00                 |          |          | 1071+      | DC                                          | XL1'00'                 |                                     |  |
| 0000183A | C3C1D3C7 40404040  |          |          | 1072+      | DC                                          | CL8'CALG'               | instruction name                    |  |
| 00001848 | 00000000 000018D8  |          |          | 1073+01A10 | DC                                          | AD(RE10+8)              | address of op1 source               |  |
| 00001850 | 00000000 000018E0  |          |          | 1074+02A10 | DC                                          | AD(RE10+16)             | address of op2 source               |  |
| 00001858 | 00000000 000018E8  |          |          | 1075+03A10 | DC                                          | AD(RE10+24)             | address of op3/R3 source            |  |
| 00001860 | 00000008           |          |          | 1076+      | DC                                          | A(8)                    | result length                       |  |
| 00001868 | 00000000 000018D0  |          |          | 1077+REA10 | DC                                          | AD(RE10)                | result address                      |  |
| 00001870 | 00000000 00000000  |          |          | 1078+      | DS                                          | FD                      | gap                                 |  |
| 00001878 | 00000000 00000000  |          |          | 1079+R3010 | DS                                          | D                       | Op3/R3 Output                       |  |
| 00001880 | 00000000 00000000  |          |          | 1080+      | DS                                          | FD                      | gap                                 |  |
| 00001888 | 00000000 00000000  |          |          | 1081+CCP10 | DS                                          | 2F                      | extract PSW after test (has CC)     |  |
| 00001890 | 00                 |          |          | 1082+CCF10 | DS                                          | X                       | extracted cc                        |  |
| 00001898 | 00000000 00000000  |          |          | 1083+      | DS                                          | FD                      |                                     |  |
|          |                    |          |          | 1084+*     |                                             |                         | INSTRUCTION UNDER TEST ROUTINE      |  |
| 000018A0 |                    |          |          | 1085+X10   | DS                                          | 0F                      |                                     |  |
| 000018A0 | E310 5018 0004     |          | 00001848 | 1086+      | LG                                          | R1,01A10                | get R1 source                       |  |
| 000018A6 | E320 5020 0004     |          | 00001850 | 1087+      | LG                                          | R2,02A10                | get R2 source                       |  |
| 000018AC | E330 5028 0004     |          | 00001858 | 1088+      | LG                                          | R3,03A10                | get R3 source address               |  |
| 000018B2 | E333 0000 0004     |          | 00000000 | 1089+      | LG                                          | R3,0(R3)                | get R3 source value                 |  |
| 000018B8 |                    |          |          | 1092+      | DS                                          | 0H                      |                                     |  |
| 000018B8 | C8371000 2000      |          |          | 1093+      | DC                                          | XL6'C83710002000'       | CALG r3,0(R1),0(R2)                 |  |
| 000018BE | E330 5048 0024     |          | 00001878 | 1095+      | STG                                         | R3,R3010                |                                     |  |
| 000018C4 | B98D 0040          |          |          | 1096+      | EPSW                                        | R4,R0                   | extract psw                         |  |
| 000018C8 | 5040 5058          |          | 00001888 | 1097+      | ST                                          | R4,CCP10                | to save CC                          |  |
| 000018CC | 07FB               |          |          | 1098+      | BR                                          | R11                     |                                     |  |
| 000018D0 |                    |          |          | 1099+RE10  | DC                                          | 0F                      |                                     |  |
| 000018D0 |                    |          |          | 1100+      | DROP                                        | R5                      |                                     |  |
| 000018D0 | 00000001 000000FF  |          |          | 1101       | DC                                          | XL8'00000001 000000FF'  | result                              |  |
| 000018D8 | 00000001 000000FF  |          |          | 1102       | DC                                          | XL8'00000001 000000FF'  | op1                                 |  |
| 000018E0 | 00000004 000000FF  |          |          | 1103       | DC                                          | XL8'00000004 000000FF'  | op2                                 |  |
| 000018E8 | FF000000 00000000  |          |          | 1104       | DC                                          | XL8'FF000000 00000000'  | r3                                  |  |





| LOC      | OBJECT CODE         | ADDR1    | ADDR2    | STMT       |       |                          |                                     |
|----------|---------------------|----------|----------|------------|-------|--------------------------|-------------------------------------|
| 000019E0 | 00000008            |          |          | 1162+      | DC    | A(8)                     | result length                       |
| 000019E8 | 00000000 00001A50   |          |          | 1163+REA12 | DC    | AD(RE12)                 | result address                      |
| 000019F0 | 00000000 00000000   |          |          | 1164+      | DS    | FD                       | gap                                 |
| 000019F8 | 00000000 00000000   |          |          | 1165+R3012 | DS    | D                        | Op3/R3 Output                       |
| 00001A00 | 00000000 00000000   |          |          | 1166+      | DS    | FD                       | gap                                 |
| 00001A08 | 00000000 00000000   |          |          | 1167+CCP12 | DS    | 2F                       | extract PSW after test (has CC)     |
| 00001A10 | 00                  |          |          | 1168+CCF12 | DS    | X                        | extracted cc                        |
| 00001A18 | 00000000 00000000   |          |          | 1169+      | DS    | FD                       |                                     |
|          |                     |          |          | 1170+*     |       |                          | INSTRUCTION UNDER TEST ROUTINE      |
| 00001A20 |                     |          |          | 1171+X12   | DS    | 0F                       |                                     |
| 00001A20 | E310 5018 0004      |          | 000019C8 | 1172+      | LG    | R1,01A12                 | get R1 source                       |
| 00001A26 | E320 5020 0004      |          | 000019D0 | 1173+      | LG    | R2,02A12                 | get R2 source                       |
| 00001A2C | E330 5028 0004      |          | 000019D8 | 1174+      | LG    | R3,03A12                 | get R3 source address               |
| 00001A32 | E333 0000 0004      |          | 00000000 | 1175+      | LG    | R3,0(R3)                 | get R3 source value                 |
| 00001A38 |                     |          |          | 1178+      | DS    | 0H                       |                                     |
| 00001A38 | C8371000 2000       |          |          | 1179+      | DC    | XL6'C83710002000'        | CALG r3,0(R1),0(R2)                 |
| 00001A3E | E330 5048 0024      |          | 000019F8 | 1181+      | STG   | R3,R3012                 |                                     |
| 00001A44 | B98D 0040           |          |          | 1182+      | EPSW  | R4,R0                    | extract psw                         |
| 00001A48 | 5040 5058           |          | 00001A08 | 1183+      | ST    | R4,CCP12                 | to save CC                          |
| 00001A4C | 07FB                |          |          | 1184+      | BR    | R11                      |                                     |
| 00001A50 |                     |          |          | 1185+RE12  | DC    | 0F                       |                                     |
| 00001A50 |                     |          |          | 1186+      | DROP  | R5                       |                                     |
| 00001A50 | CCCC0000 AAAAAAAAAA |          |          | 1187       | DC    | XL8'CCCC0000 AAAAAAAAAA' | result                              |
| 00001A58 | CCCC0000 AAAAAAAAAA |          |          | 1188       | DC    | XL8'CCCC0000 AAAAAAAAAA' | op1                                 |
| 00001A60 | 99999999 BBBB BBBB  |          |          | 1189       | DC    | XL8'99999999 BBBB BBBB'  | op2                                 |
| 00001A68 | FFFFFFFF 00009999   |          |          | 1190       | DC    | XL8'FFFFFFFF 00009999'   | r3                                  |
|          |                     |          |          | 1191       |       |                          |                                     |
|          |                     |          |          | 1192 *     |       |                          |                                     |
|          |                     |          |          | 1193 *     |       |                          |                                     |
|          |                     |          |          | 1194 *     |       |                          |                                     |
|          |                     |          |          | 1195 *     |       |                          |                                     |
|          |                     |          |          | 1196       | SSF   | CALGF,0                  |                                     |
| 00001A70 |                     |          |          | 1197+      | DS    | 0FD                      |                                     |
| 00001A70 |                     | 00001A70 |          | 1198+      | USING | *,R5                     | base for test data and test routine |
| 00001A70 | 00001AE0            |          |          | 1199+T13   | DC    | A(X13)                   | address of test routine             |
| 00001A74 | 000D                |          |          | 1200+      | DC    | H'13'                    | test number                         |
| 00001A76 | 00                  |          |          | 1201+      | DC    | XL1'00'                  |                                     |
| 00001A77 | 00                  |          |          | 1202+      | DC    | HL1'0'                   | cc                                  |
| 00001A78 | 07                  |          |          | 1203+      | DC    | HL1'7'                   | cc failed mask                      |
| 00001A79 | 00                  |          |          | 1204+      | DC    | XL1'00'                  |                                     |
| 00001A7A | C3C1D3C7 C6404040   |          |          | 1205+      | DC    | CL8'CALGF'               | instruction name                    |
| 00001A88 | 00000000 00001B18   |          |          | 1206+01A13 | DC    | AD(RE13+8)               | address of op1 source               |
| 00001A90 | 00000000 00001B20   |          |          | 1207+02A13 | DC    | AD(RE13+16)              | address of op2 source               |
| 00001A98 | 00000000 00001B28   |          |          | 1208+03A13 | DC    | AD(RE13+24)              | address of op3/R3 source            |
| 00001AA0 | 00000008            |          |          | 1209+      | DC    | A(8)                     | result length                       |
| 00001AA8 | 00000000 00001B10   |          |          | 1210+REA13 | DC    | AD(RE13)                 | result address                      |
| 00001AB0 | 00000000 00000000   |          |          | 1211+      | DS    | FD                       | gap                                 |
| 00001AB8 | 00000000 00000000   |          |          | 1212+R3013 | DS    | D                        | Op3/R3 Output                       |
| 00001AC0 | 00000000 00000000   |          |          | 1213+      | DS    | FD                       | gap                                 |
| 00001AC8 | 00000000 00000000   |          |          | 1214+CCP13 | DS    | 2F                       | extract PSW after test (has CC)     |
| 00001AD0 | 00                  |          |          | 1215+CCF13 | DS    | X                        | extracted cc                        |
| 00001AD8 | 00000000 00000000   |          |          | 1216+      | DS    | FD                       |                                     |
|          |                     |          |          | 1217+*     |       |                          | INSTRUCTION UNDER TEST ROUTINE      |
| 00001AE0 |                     |          |          | 1218+X13   | DS    | 0F                       |                                     |

| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT       |       |                        |                                     |  |
|----------|-------------------|----------|----------|------------|-------|------------------------|-------------------------------------|--|
| 00001AE0 | E310 5018 0004    |          | 00001A88 | 1219+      | LG    | R1,01A13               | get R1 source                       |  |
| 00001AE6 | E320 5020 0004    |          | 00001A90 | 1220+      | LG    | R2,02A13               | get R2 source                       |  |
| 00001AEC | E330 5028 0004    |          | 00001A98 | 1221+      | LG    | R3,03A13               | get R3 source address               |  |
| 00001AF2 | E333 0000 0004    |          | 00000000 | 1222+      | LG    | R3,0(R3)               | get R3 source value                 |  |
| 00001AF8 |                   |          |          | 1225+      | DS    | 0H                     |                                     |  |
| 00001AF8 | C83F1000 2000     |          |          | 1226+      | DC    | XL6'C83F10002000'      | CALGF r3,0(R1),0(R2)                |  |
| 00001AFE | E330 5048 0024    |          | 00001AB8 | 1228+      | STG   | R3,R3013               |                                     |  |
| 00001B04 | B98D 0040         |          |          | 1229+      | EPSW  | R4,R0                  | extract psw                         |  |
| 00001B08 | 5040 5058         |          | 00001AC8 | 1230+      | ST    | R4,CCP13               | to save CC                          |  |
| 00001B0C | 07FB              |          |          | 1231+      | BR    | R11                    |                                     |  |
| 00001B10 |                   |          |          | 1232+RE13  | DC    | 0F                     |                                     |  |
| 00001B10 |                   |          |          | 1233+      | DROP  | R5                     |                                     |  |
| 00001B10 | 00000004 000000BB |          |          | 1234       | DC    | XL8'00000004 000000BB' | result                              |  |
| 00001B18 | 00000001 000000AA |          |          | 1235       | DC    | XL8'00000001 000000AA' | op1 (first 32 bits)                 |  |
| 00001B20 | 00000004 000000BB |          |          | 1236       | DC    | XL8'00000004 000000BB' | op2                                 |  |
| 00001B28 | FF000000 00000001 |          |          | 1237       | DC    | XL8'FF000000 00000001' | r3                                  |  |
|          |                   |          |          | 1238       |       |                        |                                     |  |
|          |                   |          |          | 1239       | SSF   | CALGF,0                |                                     |  |
| 00001B30 |                   |          |          | 1240+      | DS    | 0FD                    |                                     |  |
| 00001B30 |                   | 00001B30 |          | 1241+      | USING | *,R5                   | base for test data and test routine |  |
| 00001B30 | 00001BA0          |          |          | 1242+T14   | DC    | A(X14)                 | address of test routine             |  |
| 00001B34 | 000E              |          |          | 1243+      | DC    | H'14'                  | test number                         |  |
| 00001B36 | 00                |          |          | 1244+      | DC    | XL1'00'                |                                     |  |
| 00001B37 | 00                |          |          | 1245+      | DC    | HL1'0'                 | cc                                  |  |
| 00001B38 | 07                |          |          | 1246+      | DC    | HL1'7'                 | cc failed mask                      |  |
| 00001B39 | 00                |          |          | 1247+      | DC    | XL1'00'                |                                     |  |
| 00001B3A | C3C1D3C7 C6404040 |          |          | 1248+      | DC    | CL8'CALGF'             | instruction name                    |  |
| 00001B48 | 00000000 00001BD8 |          |          | 1249+01A14 | DC    | AD(RE14+8)             | address of op1 source               |  |
| 00001B50 | 00000000 00001BE0 |          |          | 1250+02A14 | DC    | AD(RE14+16)            | address of op2 source               |  |
| 00001B58 | 00000000 00001BE8 |          |          | 1251+03A14 | DC    | AD(RE14+24)            | address of op3/R3 source            |  |
| 00001B60 | 00000008          |          |          | 1252+      | DC    | A(8)                   | result length                       |  |
| 00001B68 | 00000000 00001BD0 |          |          | 1253+REA14 | DC    | AD(RE14)               | result address                      |  |
| 00001B70 | 00000000 00000000 |          |          | 1254+      | DS    | FD                     | gap                                 |  |
| 00001B78 | 00000000 00000000 |          |          | 1255+R3014 | DS    | D                      | Op3/R3 Output                       |  |
| 00001B80 | 00000000 00000000 |          |          | 1256+      | DS    | FD                     | gap                                 |  |
| 00001B88 | 00000000 00000000 |          |          | 1257+CCP14 | DS    | 2F                     | extract PSW after test (has CC)     |  |
| 00001B90 | 00                |          |          | 1258+CCF14 | DS    | X                      | extracted cc                        |  |
| 00001B98 | 00000000 00000000 |          |          | 1259+      | DS    | FD                     |                                     |  |
|          |                   |          |          | 1260+*     |       |                        | INSTRUCTION UNDER TEST ROUTINE      |  |
| 00001BA0 |                   |          |          | 1261+X14   | DS    | 0F                     |                                     |  |
| 00001BA0 | E310 5018 0004    |          | 00001B48 | 1262+      | LG    | R1,01A14               | get R1 source                       |  |
| 00001BA6 | E320 5020 0004    |          | 00001B50 | 1263+      | LG    | R2,02A14               | get R2 source                       |  |
| 00001BAC | E330 5028 0004    |          | 00001B58 | 1264+      | LG    | R3,03A14               | get R3 source address               |  |
| 00001BB2 | E333 0000 0004    |          | 00000000 | 1265+      | LG    | R3,0(R3)               | get R3 source value                 |  |
| 00001BB8 |                   |          |          | 1268+      | DS    | 0H                     |                                     |  |
| 00001BB8 | C83F1000 2000     |          |          | 1269+      | DC    | XL6'C83F10002000'      | CALGF r3,0(R1),0(R2)                |  |
| 00001BBE | E330 5048 0024    |          | 00001B78 | 1271+      | STG   | R3,R3014               |                                     |  |
| 00001BC4 | B98D 0040         |          |          | 1272+      | EPSW  | R4,R0                  | extract psw                         |  |
| 00001BC8 | 5040 5058         |          | 00001B88 | 1273+      | ST    | R4,CCP14               | to save CC                          |  |
| 00001BCC | 07FB              |          |          | 1274+      | BR    | R11                    |                                     |  |
| 00001BD0 |                   |          |          | 1275+RE14  | DC    | 0F                     |                                     |  |
| 00001BD0 |                   |          |          | 1276+      | DROP  | R5                     |                                     |  |

| LOC      | OBJECT CODE    | ADDR1    | ADDR2    | STMT       |                                             |                   |           |                                     |
|----------|----------------|----------|----------|------------|---------------------------------------------|-------------------|-----------|-------------------------------------|
| 00001BD0 | A0000004       | BBBBBBBB |          | 1277       | DC                                          | XL8'A0000004      | BBBBBBBB' | result                              |
| 00001BD8 | 000000CC       | AAAAAAAA |          | 1278       | DC                                          | XL8'000000CC      | AAAAAAAA' | op1 (first 32 bits)                 |
| 00001BE0 | A0000004       | BBBBBBBB |          | 1279       | DC                                          | XL8'A0000004      | BBBBBBBB' | op2                                 |
| 00001BE8 | FFFFFFFF       | 000000CC |          | 1280       | DC                                          | XL8'FFFFFFFF      | 000000CC' | r3                                  |
|          |                |          |          | 1281       |                                             |                   |           |                                     |
|          |                |          |          | 1282       | SSF                                         | CALGF,0           |           |                                     |
| 00001BF0 |                |          |          | 1283+      | DS                                          | 0FD               |           |                                     |
| 00001BF0 |                | 00001BF0 |          | 1284+      | USING                                       | *,R5              |           | base for test data and test routine |
| 00001BF0 | 00001C60       |          |          | 1285+T15   | DC                                          | A(X15)            |           | address of test routine             |
| 00001BF4 | 000F           |          |          | 1286+      | DC                                          | H'15'             |           | test number                         |
| 00001BF6 | 00             |          |          | 1287+      | DC                                          | XL1'00'           |           |                                     |
| 00001BF7 | 00             |          |          | 1288+      | DC                                          | HL1'0'            |           | cc                                  |
| 00001BF8 | 07             |          |          | 1289+      | DC                                          | HL1'7'            |           | cc failed mask                      |
| 00001BF9 | 00             |          |          | 1290+      | DC                                          | XL1'00'           |           |                                     |
| 00001BFA | C3C1D3C7       | C6404040 |          | 1291+      | DC                                          | CL8'CALGF'        |           | instruction name                    |
| 00001C08 | 00000000       | 00001C98 |          | 1292+01A15 | DC                                          | AD(RE15+8)        |           | address of op1 source               |
| 00001C10 | 00000000       | 00001CA0 |          | 1293+02A15 | DC                                          | AD(RE15+16)       |           | address of op2 source               |
| 00001C18 | 00000000       | 00001CA8 |          | 1294+03A15 | DC                                          | AD(RE15+24)       |           | address of op3/R3 source            |
| 00001C20 | 00000008       |          |          | 1295+      | DC                                          | A(8)              |           | result length                       |
| 00001C28 | 00000000       | 00001C90 |          | 1296+REA15 | DC                                          | AD(RE15)          |           | result address                      |
| 00001C30 | 00000000       | 00000000 |          | 1297+      | DS                                          | FD                |           | gap                                 |
| 00001C38 | 00000000       | 00000000 |          | 1298+R3015 | DS                                          | D                 |           | Op3/R3 Output                       |
| 00001C40 | 00000000       | 00000000 |          | 1299+      | DS                                          | FD                |           | gap                                 |
| 00001C48 | 00000000       | 00000000 |          | 1300+CCP15 | DS                                          | 2F                |           | extract PSW after test (has CC)     |
| 00001C50 | 00             |          |          | 1301+CCF15 | DS                                          | X                 |           | extracted cc                        |
| 00001C58 | 00000000       | 00000000 |          | 1302+      | DS                                          | FD                |           |                                     |
|          |                |          |          | 1303+*     |                                             |                   |           | INSTRUCTION UNDER TEST ROUTINE      |
| 00001C60 |                |          |          | 1304+X15   | DS                                          | 0F                |           |                                     |
| 00001C60 | E310 5018 0004 |          | 00001C08 | 1305+      | LG                                          | R1,01A15          |           | get R1 source                       |
| 00001C66 | E320 5020 0004 |          | 00001C10 | 1306+      | LG                                          | R2,02A15          |           | get R2 source                       |
| 00001C6C | E330 5028 0004 |          | 00001C18 | 1307+      | LG                                          | R3,03A15          |           | get R3 source address               |
| 00001C72 | E333 0000 0004 |          | 00000000 | 1308+      | LG                                          | R3,0(R3)          |           | get R3 source value                 |
|          |                |          |          |            |                                             |                   |           |                                     |
| 00001C78 |                |          |          | 1311+      | DS                                          | 0H                |           |                                     |
| 00001C78 | C83F1000 2000  |          |          | 1312+      | DC                                          | XL6'C83F10002000' |           | CALGF r3,0(R1),0(R2)                |
|          |                |          |          |            |                                             |                   |           |                                     |
| 00001C7E | E330 5048 0024 |          | 00001C38 | 1314+      | STG                                         | R3,R3015          |           |                                     |
| 00001C84 | B98D 0040      |          |          | 1315+      | EPSW                                        | R4,R0             |           | extract psw                         |
| 00001C88 | 5040 5058      |          | 00001C48 | 1316+      | ST                                          | R4,CCP15          |           | to save CC                          |
| 00001C8C | 07FB           |          |          | 1317+      | BR                                          | R11               |           |                                     |
| 00001C90 |                |          |          | 1318+RE15  | DC                                          | 0F                |           |                                     |
| 00001C90 |                |          |          | 1319+      | DROP                                        | R5                |           |                                     |
| 00001C90 | 99999999       | BBBBBBBB |          | 1320       | DC                                          | XL8'99999999      | BBBBBBBB' | result                              |
| 00001C98 | 0000DDDD       | AAAAAAAA |          | 1321       | DC                                          | XL8'0000DDDD      | AAAAAAAA' | op1 (first 32 bits)                 |
| 00001CA0 | 99999999       | BBBBBBBB |          | 1322       | DC                                          | XL8'99999999      | BBBBBBBB' | op2                                 |
| 00001CA8 | FFFFFFFF       | 0000DDDD |          | 1323       | DC                                          | XL8'FFFFFFFF      | 0000DDDD' | r3                                  |
|          |                |          |          | 1324       |                                             |                   |           |                                     |
|          |                |          |          | 1325       | * CC = 1 (not equal) if r3 != op1; r3 = op1 |                   |           |                                     |
|          |                |          |          | 1326       | SSF                                         | CALGF,1           |           |                                     |
| 00001CB0 |                |          |          | 1327+      | DS                                          | 0FD               |           |                                     |
| 00001CB0 |                | 00001CB0 |          | 1328+      | USING                                       | *,R5              |           | base for test data and test routine |
| 00001CB0 | 00001D20       |          |          | 1329+T16   | DC                                          | A(X16)            |           | address of test routine             |
| 00001CB4 | 0010           |          |          | 1330+      | DC                                          | H'16'             |           | test number                         |
| 00001CB6 | 00             |          |          | 1331+      | DC                                          | XL1'00'           |           |                                     |
| 00001CB7 | 01             |          |          | 1332+      | DC                                          | HL1'1'            |           | cc                                  |
| 00001CB8 | 0B             |          |          | 1333+      | DC                                          | HL1'11'           |           | cc failed mask                      |



| LOC      | OBJECT CODE       | ADDR1    | ADDR2    | STMT       |       |                        |                                     |
|----------|-------------------|----------|----------|------------|-------|------------------------|-------------------------------------|
| 00001CB9 | 00                |          |          | 1334+      | DC    | XL1'00'                |                                     |
| 00001CBA | C3C1D3C7 C6404040 |          |          | 1335+      | DC    | CL8'CALGF'             | instruction name                    |
| 00001CC8 | 00000000 00001D58 |          |          | 1336+01A16 | DC    | AD(RE16+8)             | address of op1 source               |
| 00001CD0 | 00000000 00001D60 |          |          | 1337+02A16 | DC    | AD(RE16+16)            | address of op2 source               |
| 00001CD8 | 00000000 00001D68 |          |          | 1338+03A16 | DC    | AD(RE16+24)            | address of op3/R3 source            |
| 00001CE0 | 00000008          |          |          | 1339+      | DC    | A(8)                   | result length                       |
| 00001CE8 | 00000000 00001D50 |          |          | 1340+REA16 | DC    | AD(RE16)               | result address                      |
| 00001CF0 | 00000000 00000000 |          |          | 1341+      | DS    | FD                     | gap                                 |
| 00001CF8 | 00000000 00000000 |          |          | 1342+R3016 | DS    | D                      | Op3/R3 Output                       |
| 00001D00 | 00000000 00000000 |          |          | 1343+      | DS    | FD                     | gap                                 |
| 00001D08 | 00000000 00000000 |          |          | 1344+CCP16 | DS    | 2F                     | extract PSW after test (has CC)     |
| 00001D10 | 00                |          |          | 1345+CCF16 | DS    | X                      | extracted cc                        |
| 00001D18 | 00000000 00000000 |          |          | 1346+      | DS    | FD                     |                                     |
|          |                   |          |          | 1347+*     |       |                        | INSTRUCTION UNDER TEST ROUTINE      |
| 00001D20 |                   |          |          | 1348+X16   | DS    | 0F                     |                                     |
| 00001D20 | E310 5018 0004    |          | 00001CC8 | 1349+      | LG    | R1,01A16               | get R1 source                       |
| 00001D26 | E320 5020 0004    |          | 00001CD0 | 1350+      | LG    | R2,02A16               | get R2 source                       |
| 00001D2C | E330 5028 0004    |          | 00001CD8 | 1351+      | LG    | R3,03A16               | get R3 source address               |
| 00001D32 | E333 0000 0004    |          | 00000000 | 1352+      | LG    | R3,0(R3)               | get R3 source value                 |
| 00001D38 |                   |          |          | 1355+      | DS    | 0H                     |                                     |
| 00001D38 | C83F1000 2000     |          |          | 1356+      | DC    | XL6'C83F10002000'      | CALGF r3,0(R1),0(R2)                |
| 00001D3E | E330 5048 0024    |          | 00001CF8 | 1358+      | STG   | R3,R3016               |                                     |
| 00001D44 | B98D 0040         |          |          | 1359+      | EPSW  | R4,R0                  | extract psw                         |
| 00001D48 | 5040 5058         |          | 00001D08 | 1360+      | ST    | R4,CCP16               | to save CC                          |
| 00001D4C | 07FB              |          |          | 1361+      | BR    | R11                    |                                     |
| 00001D50 |                   |          |          | 1362+RE16  | DC    | 0F                     |                                     |
| 00001D50 |                   |          |          | 1363+      | DROP  | R5                     |                                     |
| 00001D50 | 00000000 00000001 |          |          | 1364       | DC    | XL8'00000000 00000001' | result                              |
| 00001D58 | 00000001 000000FF |          |          | 1365       | DC    | XL8'00000001 000000FF' | op1 (first 32 bits)                 |
| 00001D60 | 00000004 000000FF |          |          | 1366       | DC    | XL8'00000004 000000FF' | op2                                 |
| 00001D68 | FF0000FF 11111111 |          |          | 1367       | DC    | XL8'FF0000FF 11111111' | r3                                  |
|          |                   |          |          | 1368       |       |                        |                                     |
|          |                   |          |          | 1369       | SSF   | CALGF,1                |                                     |
| 00001D70 |                   |          |          | 1370+      | DS    | 0FD                    |                                     |
| 00001D70 |                   | 00001D70 |          | 1371+      | USING | *,R5                   | base for test data and test routine |
| 00001D70 | 00001DE0          |          |          | 1372+T17   | DC    | A(X17)                 | address of test routine             |
| 00001D74 | 0011              |          |          | 1373+      | DC    | H'17'                  | test number                         |
| 00001D76 | 00                |          |          | 1374+      | DC    | XL1'00'                |                                     |
| 00001D77 | 01                |          |          | 1375+      | DC    | HL1'1'                 | cc                                  |
| 00001D78 | 0B                |          |          | 1376+      | DC    | HL1'11'                | cc failed mask                      |
| 00001D79 | 00                |          |          | 1377+      | DC    | XL1'00'                |                                     |
| 00001D7A | C3C1D3C7 C6404040 |          |          | 1378+      | DC    | CL8'CALGF'             | instruction name                    |
| 00001D88 | 00000000 00001E18 |          |          | 1379+01A17 | DC    | AD(RE17+8)             | address of op1 source               |
| 00001D90 | 00000000 00001E20 |          |          | 1380+02A17 | DC    | AD(RE17+16)            | address of op2 source               |
| 00001D98 | 00000000 00001E28 |          |          | 1381+03A17 | DC    | AD(RE17+24)            | address of op3/R3 source            |
| 00001DA0 | 00000008          |          |          | 1382+      | DC    | A(8)                   | result length                       |
| 00001DA8 | 00000000 00001E10 |          |          | 1383+REA17 | DC    | AD(RE17)               | result address                      |
| 00001DB0 | 00000000 00000000 |          |          | 1384+      | DS    | FD                     | gap                                 |
| 00001DB8 | 00000000 00000000 |          |          | 1385+R3017 | DS    | D                      | Op3/R3 Output                       |
| 00001DC0 | 00000000 00000000 |          |          | 1386+      | DS    | FD                     | gap                                 |
| 00001DC8 | 00000000 00000000 |          |          | 1387+CCP17 | DS    | 2F                     | extract PSW after test (has CC)     |
| 00001DD0 | 00                |          |          | 1388+CCF17 | DS    | X                      | extracted cc                        |
| 00001DD8 | 00000000 00000000 |          |          | 1389+      | DS    | FD                     |                                     |
|          |                   |          |          | 1390+*     |       |                        | INSTRUCTION UNDER TEST ROUTINE      |

| LOC      | OBJECT CODE         | ADDR1    | ADDR2    | STMT       |       |                          |                                     |  |
|----------|---------------------|----------|----------|------------|-------|--------------------------|-------------------------------------|--|
| 00001DE0 |                     |          |          | 1391+X17   | DS    | 0F                       |                                     |  |
| 00001DE0 | E310 5018 0004      |          | 00001D88 | 1392+      | LG    | R1,01A17                 | get R1 source                       |  |
| 00001DE6 | E320 5020 0004      |          | 00001D90 | 1393+      | LG    | R2,02A17                 | get R2 source                       |  |
| 00001DEC | E330 5028 0004      |          | 00001D98 | 1394+      | LG    | R3,03A17                 | get R3 source address               |  |
| 00001DF2 | E333 0000 0004      |          | 00000000 | 1395+      | LG    | R3,0(R3)                 | get R3 source value                 |  |
| 00001DF8 |                     |          |          | 1398+      | DS    | 0H                       |                                     |  |
| 00001DF8 | C83F1000 2000       |          |          | 1399+      | DC    | XL6'C83F10002000'        | CALGF r3,0(R1),0(R2)                |  |
| 00001DFE | E330 5048 0024      |          | 00001DB8 | 1401+      | STG   | R3,R3017                 |                                     |  |
| 00001E04 | B98D 0040           |          |          | 1402+      | EPSW  | R4,R0                    | extract psw                         |  |
| 00001E08 | 5040 5058           |          | 00001DC8 | 1403+      | ST    | R4,CCP17                 | to save CC                          |  |
| 00001E0C | 07FB                |          |          | 1404+      | BR    | R11                      |                                     |  |
| 00001E10 |                     |          |          | 1405+RE17  | DC    | 0F                       |                                     |  |
| 00001E10 |                     |          |          | 1406+      | DROP  | R5                       |                                     |  |
| 00001E10 | 00000000 00000DDD   |          |          | 1407       | DC    | XL8'00000000 00000DDD'   | result                              |  |
| 00001E18 | 00000DDD AAAAAAAAAA |          |          | 1408       | DC    | XL8'00000DDD AAAAAAAAAA' | op1 (first 32 bits)                 |  |
| 00001E20 | A0000004 BBBB BBBB  |          |          | 1409       | DC    | XL8'A0000004 BBBB BBBB'  | op2                                 |  |
| 00001E28 | FFFFFFFF 0000CCCC   |          |          | 1410       | DC    | XL8'FFFFFFFF 0000CCCC'   | r3                                  |  |
|          |                     |          |          | 1411       |       |                          |                                     |  |
|          |                     |          |          | 1412       | SSF   | CALGF,1                  |                                     |  |
| 00001E30 |                     |          |          | 1413+      | DS    | 0FD                      |                                     |  |
| 00001E30 |                     | 00001E30 |          | 1414+      | USING | *,R5                     | base for test data and test routine |  |
| 00001E30 | 00001EA0            |          |          | 1415+T18   | DC    | A(X18)                   | address of test routine             |  |
| 00001E34 | 0012                |          |          | 1416+      | DC    | H'18'                    | test number                         |  |
| 00001E36 | 00                  |          |          | 1417+      | DC    | XL1'00'                  |                                     |  |
| 00001E37 | 01                  |          |          | 1418+      | DC    | HL1'1'                   | cc                                  |  |
| 00001E38 | 0B                  |          |          | 1419+      | DC    | HL1'11'                  | cc failed mask                      |  |
| 00001E39 | 00                  |          |          | 1420+      | DC    | XL1'00'                  |                                     |  |
| 00001E3A | C3C1D3C7 C6404040   |          |          | 1421+      | DC    | CL8'CALGF'               | instruction name                    |  |
| 00001E48 | 00000000 00001ED8   |          |          | 1422+01A18 | DC    | AD(RE18+8)               | address of op1 source               |  |
| 00001E50 | 00000000 00001EE0   |          |          | 1423+02A18 | DC    | AD(RE18+16)              | address of op2 source               |  |
| 00001E58 | 00000000 00001EE8   |          |          | 1424+03A18 | DC    | AD(RE18+24)              | address of op3/R3 source            |  |
| 00001E60 | 00000008            |          |          | 1425+      | DC    | A(8)                     | result length                       |  |
| 00001E68 | 00000000 00001ED0   |          |          | 1426+REA18 | DC    | AD(RE18)                 | result address                      |  |
| 00001E70 | 00000000 00000000   |          |          | 1427+      | DS    | FD                       | gap                                 |  |
| 00001E78 | 00000000 00000000   |          |          | 1428+R3018 | DS    | D                        | Op3/R3 Output                       |  |
| 00001E80 | 00000000 00000000   |          |          | 1429+      | DS    | FD                       | gap                                 |  |
| 00001E88 | 00000000 00000000   |          |          | 1430+CCP18 | DS    | 2F                       | extract PSW after test (has CC)     |  |
| 00001E90 | 00                  |          |          | 1431+CCF18 | DS    | X                        | extracted cc                        |  |
| 00001E98 | 00000000 00000000   |          |          | 1432+      | DS    | FD                       |                                     |  |
|          |                     |          |          | 1433+*     |       |                          | INSTRUCTION UNDER TEST ROUTINE      |  |
| 00001EA0 |                     |          |          | 1434+X18   | DS    | 0F                       |                                     |  |
| 00001EA0 | E310 5018 0004      |          | 00001E48 | 1435+      | LG    | R1,01A18                 | get R1 source                       |  |
| 00001EA6 | E320 5020 0004      |          | 00001E50 | 1436+      | LG    | R2,02A18                 | get R2 source                       |  |
| 00001EAC | E330 5028 0004      |          | 00001E58 | 1437+      | LG    | R3,03A18                 | get R3 source address               |  |
| 00001EB2 | E333 0000 0004      |          | 00000000 | 1438+      | LG    | R3,0(R3)                 | get R3 source value                 |  |
| 00001EB8 |                     |          |          | 1441+      | DS    | 0H                       |                                     |  |
| 00001EB8 | C83F1000 2000       |          |          | 1442+      | DC    | XL6'C83F10002000'        | CALGF r3,0(R1),0(R2)                |  |
| 00001EBE | E330 5048 0024      |          | 00001E78 | 1444+      | STG   | R3,R3018                 |                                     |  |
| 00001EC4 | B98D 0040           |          |          | 1445+      | EPSW  | R4,R0                    | extract psw                         |  |
| 00001EC8 | 5040 5058           |          | 00001E88 | 1446+      | ST    | R4,CCP18                 | to save CC                          |  |
| 00001ECC | 07FB                |          |          | 1447+      | BR    | R11                      |                                     |  |
| 00001ED0 |                     |          |          | 1448+RE18  | DC    | 0F                       |                                     |  |

| LOC      | OBJECT   | CODE     | ADDR1 | ADDR2 | STMT   |                                       |              |           |                     |
|----------|----------|----------|-------|-------|--------|---------------------------------------|--------------|-----------|---------------------|
| 00001ED0 |          |          |       |       | 1449+  | DROP                                  | R5           |           |                     |
| 00001ED0 | 00000000 | 0000CCCC |       |       | 1450   | DC                                    | XL8'00000000 | 0000CCCC' | result              |
| 00001ED8 | 0000CCCC | AAAAAAAA |       |       | 1451   | DC                                    | XL8'0000CCCC | AAAAAAAA' | op1 (first 32 bits) |
| 00001EE0 | 99999999 | BBBBBBBB |       |       | 1452   | DC                                    | XL8'99999999 | BBBBBBBB' | op2                 |
| 00001EE8 | FFFFFFFF | 00009999 |       |       | 1453   | DC                                    | XL8'FFFFFFFF | 00009999' | r3                  |
|          |          |          |       |       | 1454   |                                       |              |           |                     |
|          |          |          |       |       | 1455   |                                       |              |           |                     |
| 00001EF0 | 00000000 |          |       |       | 1456   | DC                                    | F'0'         |           | END OF TABLE        |
| 00001EF4 | 00000000 |          |       |       | 1457   | DC                                    | F'0'         |           |                     |
|          |          |          |       |       | 1458 * |                                       |              |           |                     |
|          |          |          |       |       | 1459 * | table of pointers to individual tests |              |           |                     |
|          |          |          |       |       | 1460 * |                                       |              |           |                     |
| 00001EF8 |          |          |       |       | 1461   | CFFTESTS                              | DS           | 0F        |                     |
|          |          |          |       |       | 1462   |                                       | PTTABLE      |           |                     |
| 00001EF8 |          |          |       |       | 1463+  | TTABLE                                | DS           | 0F        |                     |
| 00001EF8 | 00001170 |          |       |       | 1464+  |                                       | DC           | A(T1)     | address of test     |
| 00001EFC | 00001230 |          |       |       | 1465+  |                                       | DC           | A(T2)     | address of test     |
| 00001F00 | 000012F0 |          |       |       | 1466+  |                                       | DC           | A(T3)     | address of test     |
| 00001F04 | 000013B0 |          |       |       | 1467+  |                                       | DC           | A(T4)     | address of test     |
| 00001F08 | 00001470 |          |       |       | 1468+  |                                       | DC           | A(T5)     | address of test     |
| 00001F0C | 00001530 |          |       |       | 1469+  |                                       | DC           | A(T6)     | address of test     |
| 00001F10 | 000015F0 |          |       |       | 1470+  |                                       | DC           | A(T7)     | address of test     |
| 00001F14 | 000016B0 |          |       |       | 1471+  |                                       | DC           | A(T8)     | address of test     |
| 00001F18 | 00001770 |          |       |       | 1472+  |                                       | DC           | A(T9)     | address of test     |
| 00001F1C | 00001830 |          |       |       | 1473+  |                                       | DC           | A(T10)    | address of test     |
| 00001F20 | 000018F0 |          |       |       | 1474+  |                                       | DC           | A(T11)    | address of test     |
| 00001F24 | 000019B0 |          |       |       | 1475+  |                                       | DC           | A(T12)    | address of test     |
| 00001F28 | 00001A70 |          |       |       | 1476+  |                                       | DC           | A(T13)    | address of test     |
| 00001F2C | 00001B30 |          |       |       | 1477+  |                                       | DC           | A(T14)    | address of test     |
| 00001F30 | 00001BF0 |          |       |       | 1478+  |                                       | DC           | A(T15)    | address of test     |
| 00001F34 | 00001CB0 |          |       |       | 1479+  |                                       | DC           | A(T16)    | address of test     |
| 00001F38 | 00001D70 |          |       |       | 1480+  |                                       | DC           | A(T17)    | address of test     |
| 00001F3C | 00001E30 |          |       |       | 1481+  |                                       | DC           | A(T18)    | address of test     |
|          |          |          |       |       | 1482+* |                                       |              |           |                     |
| 00001F40 | 00000000 |          |       |       | 1483+  |                                       | DC           | A(0)      | END OF TABLE        |
| 00001F44 | 00000000 |          |       |       | 1484+  |                                       | DC           | A(0)      |                     |

| LOC | OBJECT CODE | ADDR1    | ADDR2    | STMT     |                    |    |  |
|-----|-------------|----------|----------|----------|--------------------|----|--|
|     |             |          |          | 1486     | *****              |    |  |
|     |             |          |          | 1487     | * Register equates |    |  |
|     |             |          |          | 1488     | *****              |    |  |
|     |             | 00000000 | 00000001 | 1490 R0  | EQU                | 0  |  |
|     |             | 00000001 | 00000001 | 1491 R1  | EQU                | 1  |  |
|     |             | 00000002 | 00000001 | 1492 R2  | EQU                | 2  |  |
|     |             | 00000003 | 00000001 | 1493 R3  | EQU                | 3  |  |
|     |             | 00000004 | 00000001 | 1494 R4  | EQU                | 4  |  |
|     |             | 00000005 | 00000001 | 1495 R5  | EQU                | 5  |  |
|     |             | 00000006 | 00000001 | 1496 R6  | EQU                | 6  |  |
|     |             | 00000007 | 00000001 | 1497 R7  | EQU                | 7  |  |
|     |             | 00000008 | 00000001 | 1498 R8  | EQU                | 8  |  |
|     |             | 00000009 | 00000001 | 1499 R9  | EQU                | 9  |  |
|     |             | 0000000A | 00000001 | 1500 R10 | EQU                | 10 |  |
|     |             | 0000000B | 00000001 | 1501 R11 | EQU                | 11 |  |
|     |             | 0000000C | 00000001 | 1502 R12 | EQU                | 12 |  |
|     |             | 0000000D | 00000001 | 1503 R13 | EQU                | 13 |  |
|     |             | 0000000E | 00000001 | 1504 R14 | EQU                | 14 |  |
|     |             | 0000000F | 00000001 | 1505 R15 | EQU                | 15 |  |
|     |             |          |          |          |                    |    |  |
|     |             |          |          |          |                    |    |  |
|     |             |          |          | 1507     | *****              |    |  |
|     |             |          |          | 1508     | * Register equates |    |  |
|     |             |          |          | 1509     | *****              |    |  |
|     |             | 00000000 | 00000001 | 1511 V0  | EQU                | 0  |  |
|     |             | 00000001 | 00000001 | 1512 V1  | EQU                | 1  |  |
|     |             | 00000002 | 00000001 | 1513 V2  | EQU                | 2  |  |
|     |             | 00000003 | 00000001 | 1514 V3  | EQU                | 3  |  |
|     |             | 00000004 | 00000001 | 1515 V4  | EQU                | 4  |  |
|     |             | 00000005 | 00000001 | 1516 V5  | EQU                | 5  |  |
|     |             | 00000006 | 00000001 | 1517 V6  | EQU                | 6  |  |
|     |             | 00000007 | 00000001 | 1518 V7  | EQU                | 7  |  |
|     |             | 00000008 | 00000001 | 1519 V8  | EQU                | 8  |  |
|     |             | 00000009 | 00000001 | 1520 V9  | EQU                | 9  |  |
|     |             | 0000000A | 00000001 | 1521 V10 | EQU                | 10 |  |
|     |             | 0000000B | 00000001 | 1522 V11 | EQU                | 11 |  |
|     |             | 0000000C | 00000001 | 1523 V12 | EQU                | 12 |  |
|     |             | 0000000D | 00000001 | 1524 V13 | EQU                | 13 |  |
|     |             | 0000000E | 00000001 | 1525 V14 | EQU                | 14 |  |
|     |             | 0000000F | 00000001 | 1526 V15 | EQU                | 15 |  |
|     |             | 00000010 | 00000001 | 1527 V16 | EQU                | 16 |  |
|     |             | 00000011 | 00000001 | 1528 V17 | EQU                | 17 |  |
|     |             | 00000012 | 00000001 | 1529 V18 | EQU                | 18 |  |
|     |             | 00000013 | 00000001 | 1530 V19 | EQU                | 19 |  |
|     |             | 00000014 | 00000001 | 1531 V20 | EQU                | 20 |  |
|     |             | 00000015 | 00000001 | 1532 V21 | EQU                | 21 |  |



| SYMBOL    | TYPE | VALUE    | LENGTH | DEFN | REFERENCES |     |     |     |     |     |     |     |  |  |
|-----------|------|----------|--------|------|------------|-----|-----|-----|-----|-----|-----|-----|--|--|
| BEGIN     | I    | 00000200 | 2      | 158  | 123        | 154 | 155 | 156 |     |     |     |     |  |  |
| CC        | U    | 00000007 | 1      | 497  | 256        |     |     |     |     |     |     |     |  |  |
| CCF1      | X    | 000011D0 | 1      | 689  |            |     |     |     |     |     |     |     |  |  |
| CCF10     | X    | 00001890 | 1      | 1082 |            |     |     |     |     |     |     |     |  |  |
| CCF11     | X    | 00001950 | 1      | 1125 |            |     |     |     |     |     |     |     |  |  |
| CCF12     | X    | 00001A10 | 1      | 1168 |            |     |     |     |     |     |     |     |  |  |
| CCF13     | X    | 00001AD0 | 1      | 1215 |            |     |     |     |     |     |     |     |  |  |
| CCF14     | X    | 00001B90 | 1      | 1258 |            |     |     |     |     |     |     |     |  |  |
| CCF15     | X    | 00001C50 | 1      | 1301 |            |     |     |     |     |     |     |     |  |  |
| CCF16     | X    | 00001D10 | 1      | 1345 |            |     |     |     |     |     |     |     |  |  |
| CCF17     | X    | 00001DD0 | 1      | 1388 |            |     |     |     |     |     |     |     |  |  |
| CCF18     | X    | 00001E90 | 1      | 1431 |            |     |     |     |     |     |     |     |  |  |
| CCF2      | X    | 00001290 | 1      | 732  |            |     |     |     |     |     |     |     |  |  |
| CCF3      | X    | 00001350 | 1      | 775  |            |     |     |     |     |     |     |     |  |  |
| CCF4      | X    | 00001410 | 1      | 819  |            |     |     |     |     |     |     |     |  |  |
| CCF5      | X    | 000014D0 | 1      | 862  |            |     |     |     |     |     |     |     |  |  |
| CCF6      | X    | 00001590 | 1      | 905  |            |     |     |     |     |     |     |     |  |  |
| CCF7      | X    | 00001650 | 1      | 952  |            |     |     |     |     |     |     |     |  |  |
| CCF8      | X    | 00001710 | 1      | 995  |            |     |     |     |     |     |     |     |  |  |
| CCF9      | X    | 000017D0 | 1      | 1038 |            |     |     |     |     |     |     |     |  |  |
| CCFOUND   | X    | 00000060 | 1      | 511  | 244        | 264 |     |     |     |     |     |     |  |  |
| CCMASK    | U    | 00000008 | 1      | 498  | 221        |     |     |     |     |     |     |     |  |  |
| CCMSG     | U    | 0000031E | 1      | 238  | 233        |     |     |     |     |     |     |     |  |  |
| CCP1      | F    | 000011C8 | 4      | 688  | 704        |     |     |     |     |     |     |     |  |  |
| CCP10     | F    | 00001888 | 4      | 1081 | 1097       |     |     |     |     |     |     |     |  |  |
| CCP11     | F    | 00001948 | 4      | 1124 | 1140       |     |     |     |     |     |     |     |  |  |
| CCP12     | F    | 00001A08 | 4      | 1167 | 1183       |     |     |     |     |     |     |     |  |  |
| CCP13     | F    | 00001AC8 | 4      | 1214 | 1230       |     |     |     |     |     |     |     |  |  |
| CCP14     | F    | 00001B88 | 4      | 1257 | 1273       |     |     |     |     |     |     |     |  |  |
| CCP15     | F    | 00001C48 | 4      | 1300 | 1316       |     |     |     |     |     |     |     |  |  |
| CCP16     | F    | 00001D08 | 4      | 1344 | 1360       |     |     |     |     |     |     |     |  |  |
| CCP17     | F    | 00001DC8 | 4      | 1387 | 1403       |     |     |     |     |     |     |     |  |  |
| CCP18     | F    | 00001E88 | 4      | 1430 | 1446       |     |     |     |     |     |     |     |  |  |
| CCP2      | F    | 00001288 | 4      | 731  | 747        |     |     |     |     |     |     |     |  |  |
| CCP3      | F    | 00001348 | 4      | 774  | 790        |     |     |     |     |     |     |     |  |  |
| CCP4      | F    | 00001408 | 4      | 818  | 834        |     |     |     |     |     |     |     |  |  |
| CCP5      | F    | 000014C8 | 4      | 861  | 877        |     |     |     |     |     |     |     |  |  |
| CCP6      | F    | 00001588 | 4      | 904  | 920        |     |     |     |     |     |     |     |  |  |
| CCP7      | F    | 00001648 | 4      | 951  | 967        |     |     |     |     |     |     |     |  |  |
| CCP8      | F    | 00001708 | 4      | 994  | 1010       |     |     |     |     |     |     |     |  |  |
| CCP9      | F    | 000017C8 | 4      | 1037 | 1053       |     |     |     |     |     |     |     |  |  |
| CCPRTEXP  | C    | 0000107E | 1      | 454  | 260        |     |     |     |     |     |     |     |  |  |
| CCPRTGOT  | C    | 0000108E | 1      | 457  | 268        |     |     |     |     |     |     |     |  |  |
| CCPRTLNE  | C    | 0000103B | 16     | 449  | 459        | 271 |     |     |     |     |     |     |  |  |
| CCPRTLNG  | U    | 00000055 | 1      | 459  | 270        |     |     |     |     |     |     |     |  |  |
| CCPRTNAME | C    | 00001068 | 8      | 452  | 252        |     |     |     |     |     |     |     |  |  |
| CCPRTNUM  | C    | 0000104B | 3      | 450  | 251        |     |     |     |     |     |     |     |  |  |
| CCPSW     | F    | 00000058 | 4      | 510  | 241        |     |     |     |     |     |     |     |  |  |
| CFFTEST   | 4    | 00000000 | 112    | 492  | 216        |     |     |     |     |     |     |     |  |  |
| CFFTESTS  | F    | 00001EF8 | 4      | 1461 | 209        |     |     |     |     |     |     |     |  |  |
| CFFTST    | J    | 00000000 | 8008   | 116  | 119        | 121 | 125 | 129 | 425 | 117 |     |     |  |  |
| CTLR0     | F    | 0000050C | 4      | 396  | 168        | 169 | 170 | 171 |     |     |     |     |  |  |
| DECNUM    | C    | 000010BC | 16     | 469  | 248        | 250 | 257 | 259 | 265 | 267 | 286 | 288 |  |  |
| EDIT      | X    | 00001090 | 18     | 464  | 249        | 258 | 266 | 287 |     |     |     |     |  |  |
| ENDTEST   | U    | 000003DE | 1      | 310  | 214        |     |     |     |     |     |     |     |  |  |
| EOJ       | I    | 000004F0 | 4      | 386  | 202        | 313 |     |     |     |     |     |     |  |  |

| SYMBOL   | TYPE | VALUE    | LENGTH | DEFN | REFERENCES |     |     |
|----------|------|----------|--------|------|------------|-----|-----|
| EOJPSW   | D    | 000004E0 | 8      | 384  | 386        |     |     |
| FAILCONT | U    | 000003CE | 1      | 300  |            |     |     |
| FAILED   | F    | 00001000 | 4      | 426  | 275        | 302 | 311 |
| FAILMSG  | U    | 000003A2 | 1      | 284  | 228        |     |     |
| FAILPSW  | D    | 000004F8 | 8      | 388  | 390        |     |     |
| FAILTEST | I    | 00000508 | 4      | 390  | 314        |     |     |
| FB0001   | F    | 00000290 | 8      | 186  | 190        | 191 | 193 |
| IMAGE    | 1    | 00000000 | 8008   | 0    |            |     |     |
| K        | U    | 00000400 | 1      | 409  | 410        | 411 | 412 |
| K64      | U    | 00010000 | 1      | 411  |            |     |     |
| MB       | U    | 00100000 | 1      | 412  |            |     |     |
| MSG      | I    | 00000428 | 4      | 346  | 201        | 329 |     |
| MSGCMD   | C    | 00000476 | 9      | 376  | 359        | 360 |     |
| MSGMSG   | C    | 0000047F | 95     | 377  | 353        | 374 | 351 |
| MSGMVC   | I    | 00000470 | 6      | 374  | 357        |     |     |
| MSGOK    | I    | 0000043E | 2      | 355  | 352        |     |     |
| MSGRET   | I    | 0000045E | 4      | 370  | 363        | 366 |     |
| MSGSAVE  | F    | 00000464 | 4      | 373  | 349        | 370 |     |
| NEXTCFF  | U    | 000002E4 | 1      | 211  | 231        | 305 |     |
| 01A1     | A    | 00001188 | 8      | 680  | 693        |     |     |
| 01A10    | A    | 00001848 | 8      | 1073 | 1086       |     |     |
| 01A11    | A    | 00001908 | 8      | 1116 | 1129       |     |     |
| 01A12    | A    | 000019C8 | 8      | 1159 | 1172       |     |     |
| 01A13    | A    | 00001A88 | 8      | 1206 | 1219       |     |     |
| 01A14    | A    | 00001B48 | 8      | 1249 | 1262       |     |     |
| 01A15    | A    | 00001C08 | 8      | 1292 | 1305       |     |     |
| 01A16    | A    | 00001CC8 | 8      | 1336 | 1349       |     |     |
| 01A17    | A    | 00001D88 | 8      | 1379 | 1392       |     |     |
| 01A18    | A    | 00001E48 | 8      | 1422 | 1435       |     |     |
| 01A2     | A    | 00001248 | 8      | 723  | 736        |     |     |
| 01A3     | A    | 00001308 | 8      | 766  | 779        |     |     |
| 01A4     | A    | 000013C8 | 8      | 810  | 823        |     |     |
| 01A5     | A    | 00001488 | 8      | 853  | 866        |     |     |
| 01A6     | A    | 00001548 | 8      | 896  | 909        |     |     |
| 01A7     | A    | 00001608 | 8      | 943  | 956        |     |     |
| 01A8     | A    | 000016C8 | 8      | 986  | 999        |     |     |
| 01A9     | A    | 00001788 | 8      | 1029 | 1042       |     |     |
| 01ADDR   | A    | 00000018 | 8      | 502  |            |     |     |
| 02A1     | A    | 00001190 | 8      | 681  | 694        |     |     |
| 02A10    | A    | 00001850 | 8      | 1074 | 1087       |     |     |
| 02A11    | A    | 00001910 | 8      | 1117 | 1130       |     |     |
| 02A12    | A    | 000019D0 | 8      | 1160 | 1173       |     |     |
| 02A13    | A    | 00001A90 | 8      | 1207 | 1220       |     |     |
| 02A14    | A    | 00001B50 | 8      | 1250 | 1263       |     |     |
| 02A15    | A    | 00001C10 | 8      | 1293 | 1306       |     |     |
| 02A16    | A    | 00001CD0 | 8      | 1337 | 1350       |     |     |
| 02A17    | A    | 00001D90 | 8      | 1380 | 1393       |     |     |
| 02A18    | A    | 00001E50 | 8      | 1423 | 1436       |     |     |
| 02A2     | A    | 00001250 | 8      | 724  | 737        |     |     |
| 02A3     | A    | 00001310 | 8      | 767  | 780        |     |     |
| 02A4     | A    | 000013D0 | 8      | 811  | 824        |     |     |
| 02A5     | A    | 00001490 | 8      | 854  | 867        |     |     |
| 02A6     | A    | 00001550 | 8      | 897  | 910        |     |     |
| 02A7     | A    | 00001610 | 8      | 944  | 957        |     |     |
| 02A8     | A    | 000016D0 | 8      | 987  | 1000       |     |     |
| 02A9     | A    | 00001790 | 8      | 1030 | 1043       |     |     |







| SYMBOL   | TYPE | VALUE    | LENGTH | DEFN | REFERENCES |     |  |
|----------|------|----------|--------|------|------------|-----|--|
| REA4     | A    | 000013E8 | 8      | 814  |            |     |  |
| REA5     | A    | 000014A8 | 8      | 857  |            |     |  |
| REA6     | A    | 00001568 | 8      | 900  |            |     |  |
| REA7     | A    | 00001628 | 8      | 947  |            |     |  |
| REA8     | A    | 000016E8 | 8      | 990  |            |     |  |
| REA9     | A    | 000017A8 | 8      | 1033 |            |     |  |
| READDR   | A    | 00000038 | 8      | 506  | 226        |     |  |
| REG2LOW  | U    | 000000DD | 1      | 416  |            |     |  |
| REG2PATT | U    | AABBCCDD | 1      | 415  |            |     |  |
| RELEN    | A    | 00000030 | 4      | 505  |            |     |  |
| RPTDWSAV | D    | 00000418 | 8      | 339  | 328        | 330 |  |
| RPTERROR | I    | 000003EC | 4      | 323  | 272        | 295 |  |
| RPTSAVE  | F    | 0000040C | 4      | 336  | 323        | 333 |  |
| RPTSVR5  | F    | 00000410 | 4      | 337  | 324        | 332 |  |
| SKL0001  | U    | 00000058 | 1      | 183  | 199        |     |  |
| SKT0001  | C    | 0000022A | 22     | 180  | 183        | 200 |  |
| SVOLDPSW | U    | 00000140 | 0      | 119  |            |     |  |
| T1       | A    | 00001170 | 4      | 673  | 1464       |     |  |
| T10      | A    | 00001830 | 4      | 1066 | 1473       |     |  |
| T11      | A    | 000018F0 | 4      | 1109 | 1474       |     |  |
| T12      | A    | 000019B0 | 4      | 1152 | 1475       |     |  |
| T13      | A    | 00001A70 | 4      | 1199 | 1476       |     |  |
| T14      | A    | 00001B30 | 4      | 1242 | 1477       |     |  |
| T15      | A    | 00001BF0 | 4      | 1285 | 1478       |     |  |
| T16      | A    | 00001CB0 | 4      | 1329 | 1479       |     |  |
| T17      | A    | 00001D70 | 4      | 1372 | 1480       |     |  |
| T18      | A    | 00001E30 | 4      | 1415 | 1481       |     |  |
| T2       | A    | 00001230 | 4      | 716  | 1465       |     |  |
| T3       | A    | 000012F0 | 4      | 759  | 1466       |     |  |
| T4       | A    | 000013B0 | 4      | 803  | 1467       |     |  |
| T5       | A    | 00001470 | 4      | 846  | 1468       |     |  |
| T6       | A    | 00001530 | 4      | 889  | 1469       |     |  |
| T7       | A    | 000015F0 | 4      | 936  | 1470       |     |  |
| T8       | A    | 000016B0 | 4      | 979  | 1471       |     |  |
| T9       | A    | 00001770 | 4      | 1022 | 1472       |     |  |
| TESTCC   | I    | 0000031A | 4      | 233  | 223        |     |  |
| TESTING  | F    | 00001004 | 4      | 427  |            |     |  |
| TESTREST | U    | 00000302 | 1      | 225  | 277        |     |  |
| TNUM     | H    | 00000004 | 2      | 494  | 247        | 285 |  |
| TSUB     | A    | 00000000 | 4      | 493  | 218        |     |  |
| TTABLE   | F    | 00001EF8 | 4      | 1463 |            |     |  |
| V0       | U    | 00000000 | 1      | 1511 |            |     |  |
| V1       | U    | 00000001 | 1      | 1512 |            |     |  |
| V10      | U    | 0000000A | 1      | 1521 |            |     |  |
| V11      | U    | 0000000B | 1      | 1522 |            |     |  |
| V12      | U    | 0000000C | 1      | 1523 |            |     |  |
| V13      | U    | 0000000D | 1      | 1524 |            |     |  |
| V14      | U    | 0000000E | 1      | 1525 |            |     |  |
| V15      | U    | 0000000F | 1      | 1526 |            |     |  |
| V16      | U    | 00000010 | 1      | 1527 |            |     |  |
| V17      | U    | 00000011 | 1      | 1528 |            |     |  |
| V18      | U    | 00000012 | 1      | 1529 |            |     |  |
| V19      | U    | 00000013 | 1      | 1530 |            |     |  |
| V1FUDGE  | X    | 00001120 | 16     | 481  |            |     |  |
| V1FUDGEb | X    | 00001130 | 16     | 482  |            |     |  |
| V1INPUT  | C    | 00001140 | 16     | 483  |            |     |  |





| DESC | SYMBOL | SIZE | POS | ADDR |
|------|--------|------|-----|------|
|------|--------|------|-----|------|

**Entry: 0**

|               |               |             |                  |                  |
|---------------|---------------|-------------|------------------|------------------|
| <b>Image</b>  | <b>IMAGE</b>  | <b>8008</b> | <b>0000-1F47</b> | <b>0000-1F47</b> |
| <b>Region</b> |               | <b>8008</b> | <b>0000-1F47</b> | <b>0000-1F47</b> |
| <b>CSECT</b>  | <b>CFFTST</b> | <b>8008</b> | <b>0000-1F47</b> | <b>0000-1F47</b> |

| STMT | FILE NAME |
|------|-----------|
|------|-----------|

```
1 /devstor/sharedvfp/tests/cff-01-CompareAndLoad.asm
```

**\*\* NO ERRORS FOUND \*\***